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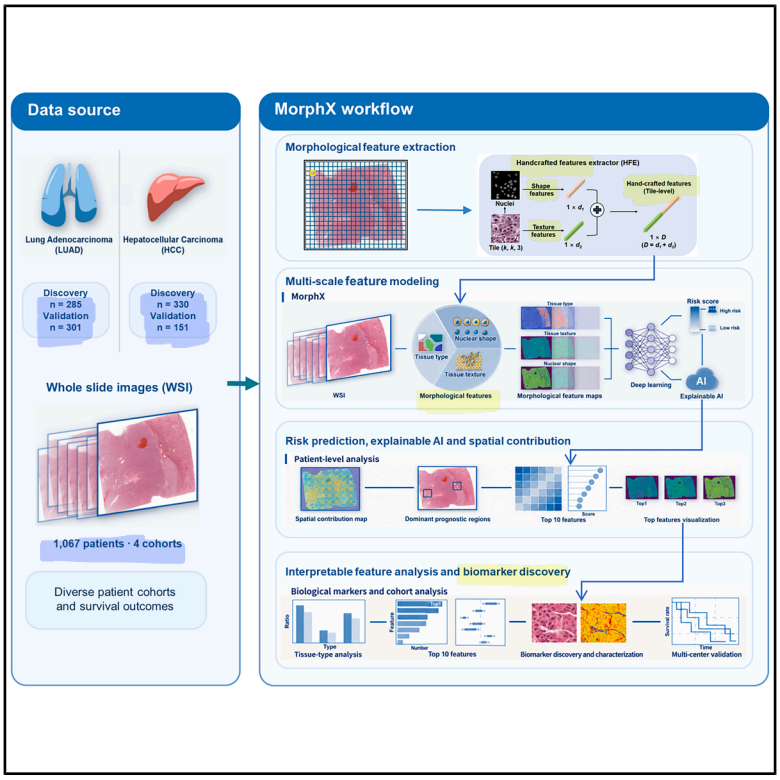
Interpretable deep learning maps spatial morphological phenotypes to HCC and LUAD survival

Graphical abstract



Highlights

- MorphX links WSI morphology to interpretable patient-level risk scores
- Multi-scale morphology captures prognostic signals across HCC and LUAD
- Attribution-guided analyses identify tissue-specific features associated with survival
- Morphology-based candidates show consistent survival associations across cohorts

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In brief

Liu et al. present MorphX, an interpretable deep learning framework that decodes cancer prognosis from whole-slide histological images and reveals morphological biomarkers. Across independent cohorts, MorphX identifies eight morphology-based biomarkers, supporting trustworthy, transparent, AI-driven precision oncology.

Liu et al., 2026, Cell Reports Medicine 7, 103126
November 17, 2026 © 2026 The Authors. Published by Elsevier Inc.
https://doi.org/10.1016/j.xcrm.2026.103126

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Interpretable deep learning maps spatial morphological phenotypes to HCC and LUAD survival

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https://doi.org/10.1016/j.xcrm.2026.103126

SUMMARY

The clinical adoption of deep learning in pathology is largely hindered by the "black box" nature of current models, which often obscure the biological basis of prognosis. Here, we introduce MorphX, an interpretable framework that predicts patient survival by explicitly decoding multi-scale morphological patterns from whole-slide images. Unlike conventional methods relying on latent features, MorphX fuses macro-scale tissue texture with micro-scale nuclear metrics to attribute risk to interpretable biological phenotypes. In a multi-center study spanning 1,067 patients with hepatocellular carcinoma (HCC) and lung adenocarcinoma (LUAD), MorphX achieves concordance indices of 0.775–0.799 in discovery cohorts. Crucially, we derive eight distinct morphology-based biomarkers—driven by texture heterogeneity in necrotic regions, lymphocyte density, and nuclear shape—that demonstrate reproducible survival associations across independent cohorts, offering a granular resolution of tumor aggressiveness that transcends traditional histological grading. By translating opaque computational predictions into validated biological insights, MorphX supports trustworthy, transparent, artificial intelligence (AI)-driven precision oncology.

INTRODUCTION

Although histological hematoxylin and eosin (H&E)-stained whole-slide images (WSIs) have long been used in clinical practice,[1] their potential to support precision medicine, particularly in prognosis prediction, remains underexplored.[2] Deep learning models, including pathology foundation models,[3,4] have demonstrated strong performance in analyzing histological WSIs.[5–8] However, the abstract deep learning features they rely on are often difficult to interpret in terms of biological meaning.[9,10] As a result, such models are rarely integrated into clinical workflows and often lack the trust of pathologists.[11] One promising direction is to integrate hand-crafted morphological features with high interpretability[12–15] into data-driven deep learning models that offer strong predictive power. Such integration facilitates modeling the complex relationships between morphological features and patient outcomes by leveraging the rich multi-scale information available in WSIs. Furthermore, with the recent progress in explainable artificial intelligence (xAI) techniques,[16] it has become possible to identify the contribution of each feature to deep learning model decisions, making it easier to interpret how hand-crafted morphological features in WSIs relate to patient prognosis.[17]

In this study, we present MorphX, an interpretable deep learning framework for WSI-based prognosis that includes

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the morphological feature model (MFM) and the multi-scale feature model (MSFM) (the architectural details are shown in Figure S1). Specifically, we construct a comprehensive set of spatially aligned WSI-level morphological feature maps that explicitly encode tissue type, tissue texture, and nuclear shape.[18,19] These maps establish an intrinsic and biologically grounded input space. Using these maps as inputs, we train the MFM to predict patient risk. We then apply an attribution-based xAI method, integrated gradients,[20] directly on the morphological feature maps to quantify feature-wise and region-wise contributions to prognosis. Guided by these attributions, we select high-contribution tiles and fuse their local features with the global features from the morphological feature maps, constructing the MSFM with improved predictive performance. We demonstrate the effectiveness and generalizability of MorphX in international multi-center cohorts of hepatocellular carcinoma (HCC) and lung adenocarcinoma (LUAD).

To support clinical decision-making, WSI-based prognostic models need to offer transparent reasoning in addition to accurate risk prediction. Concretely, clinicians need to understand which explicit morphological features drive a patient's risk estimate and where these features manifest on the slide, so that model evidence can be inspected and reconciled with established histopathological knowledge. Such feature-wise and region-wise transparency is essential for interpreting individual predictions, enabling pathologist verification, and fostering trust for clinical deployment.[11] MorphX provides such transparency by coupling explicit morphological feature maps with attribution-based xAI, enabling prognostic evidence to be traced back to concrete histological patterns.

By applying MorphX to pathological datasets, we obtain quantitative explanations of how biologically meaningful morphological features contribute to prognosis (the overview of applying MorphX for interpretable prognosis prediction is shown in Figure 1). At the patient level, MorphX assigns each morphological feature a contribution score and highlights the most prognostically informative regions within each WSI, allowing pathologists to see which patterns and areas drive an individual patient's risk assessment. At the cohort level, we summarize these feature contributions across all patients to derive a robust ranking of feature importance and characterize how these features manifest across different tissue types, revealing systematic differences between high- and low-risk patients. On the basis of these analyses, we identify potential morphology-based biomarkers associated with patient outcomes and provide a clear bridge between model predictions and clinically interpretable histological patterns.

MorphX is a computational pathology framework with tractable decision processes, enabled by xAI, which combines the interpretability of hand-crafted morphological feature maps with the predictive power of deep learning in WSIs for cancer prognosis. MorphX achieves strong prognostic performance, reveals the prognostic significance of morphological features, and identifies critical pathological regions that are closely associated with patient outcomes. These results may support personalized treatment strategies and contribute to the development of clinically meaningful morphology-based biomarkers.

RESULTS

Patient cohorts

To demonstrate the effectiveness and generalizability of MorphX across distinct histological contexts, we focus on two solid cancers with clearly different tissue architectures and morphologies: HCC and LUAD. We analyze 1,067 patients diagnosed with HCC or LUAD from international multi-center cohorts. The HCC dataset includes 330 patients from The Cancer Genome Atlas Liver Hepatocellular Carcinoma (TCGA-LIHC) and 151 patients from The First Affiliated Hospital of Kunming Medical University (KMMUFH). The LUAD dataset includes 285 patients from The Cancer Genome Atlas Lung Adenocarcinoma (TCGA-LUAD) and 301 patients from Guangdong Provincial People's Hospital (GDPH). Overall survival (OS) is defined as the time from the date of primary surgery to death or last follow-up. The four cohorts contain 112, 56, 98, and 62 OS events, respectively, and have median ages of 60, 54, 65, and 62 years. Cohort characteristics, including median observed OS time and non-missing denominators, are summarized in Table S1.

TCGA-LIHC and TCGA-LUAD serve as discovery cohorts, with KMMUFH and GDPH as independent validation cohorts. For each WSI, we extract 160 hand-crafted feature maps covering tissue texture and nuclear shape (see Figure S6A for all feature names). We combine these hand-crafted feature maps with tissue type maps containing seven types for HCC (tumor, fibrosis, necrosis, steatosis, inflammation, ductular reaction, and normal) and four types for LUAD (tumor, stroma, necrosis, and lymphocyte) to construct WSI-level morphological feature maps that comprehensively describe each patient's histological patterns (Figure 1). Published models are used for nuclear and tissue segmentation; details are provided in STAR Methods.

Development and evaluation of MorphX for prognosis

The prognostic capability of MorphX is realized through its core predictive component, MSFM, which hierarchically integrates multi-scale morphological information from WSIs to predict postoperative survival risk.

We perform 10-fold cross-validation on each discovery cohort (90% training and 10% internal test per fold). For method comparison, the checkpoint with the highest internal-test concordance index (C-index) in each fold is evaluated on the external cohort, and the same selection protocol is applied to all baselines. Ablation experiments show that using the complete morphological feature maps, which incorporate tissue texture and nuclear/cell descriptors in addition to tissue type, improves the external-test C-index from 0.4955 to 0.6724 in HCC and from 0.5439 to 0.5949 in LUAD (Table S2). Adding micro-scale features from the top 16 high-attribution tiles in MSFM yields further C-index gains, from 0.7592 to 0.7988 in TCGA-LIHC and from 0.7586 to 0.7749 in TCGA-LUAD, with corresponding external C-indices increasing from 0.6724 to 0.6761 in KMMUFH and from 0.5949 to 0.6116 in GDPH (Figures S2C and S2D). We compare MSFM with representative WSI-based prognostic models, including pathology foundation models,[21,22] multiple-instance learning (MIL) models,[23,24] and prognosis-specific WSI models[11] (Figure 2A; Table S3).

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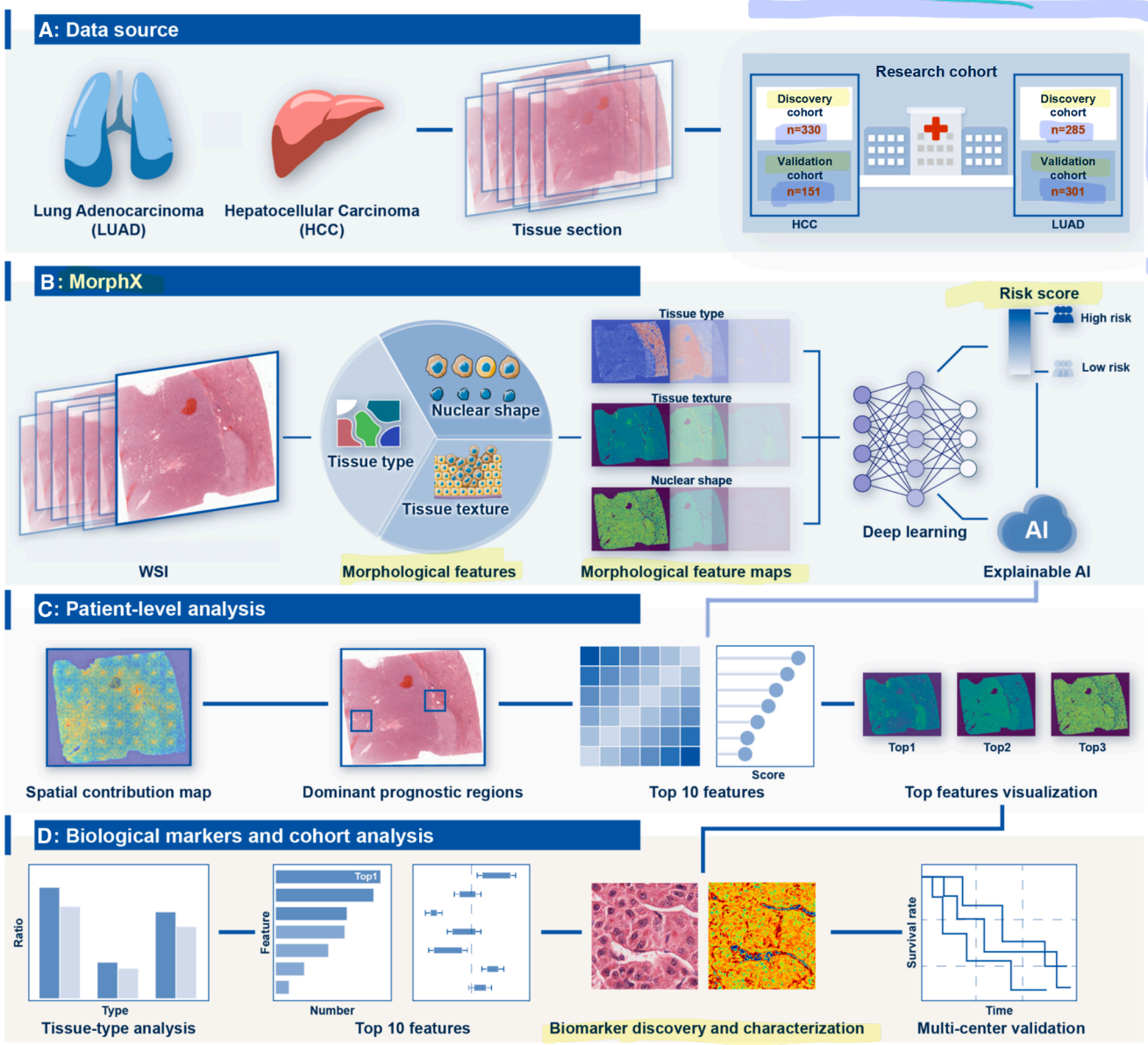

Figure 1. Overview of applying MorphX for interpretable prognosis prediction

(A–D) MorphX generates risk scores for individual patients and applies xAI methods to derive patient-level contribution distributions of morphological features (the architectural details of the MorphX framework are shown in Figure S1). Furthermore, through patient- and cohort-level analyses, we decode the associations between morphological features and prognosis and uncover potential morphology-based biomarkers. This integrated methodology yields actionable insights for precision medicine by revealing morphological determinants of clinical outcomes.

Sensitivity analyses support the complete morphology representation and the selected number of key tiles. Complete morphological maps have the highest mean external C-index in HCC (0.670 ± 0.018) and LUAD (0.626 ± 0.079), exceeding the corresponding reduced feature-family inputs (Figures S2E and S2F; Data S2). Cancer-specific normal-region sensitivity analyses reduced the external C-index to 0.514 ± 0.039 for LUAD-derived-normal-only input and 0.618 ± 0.036 after removal of mapped HCC normal tissue (Figures S2G and S2H; Data S2).

Across the evaluated models, MSFM is competitive in both discovery and external cohorts. Compared with GigaPath, MSFM increases the discovery/external C-index from 0.7668/0.6023 to 0.7988/0.6761 in HCC and from 0.7209/0.6004 to 0.7749/0.6116 in LUAD. BEiT-based model Pre-training on Histopathological image (BEPH)[25] yields 0.772 ± 0.118/0.589 ± 0.026 in HCC and 0.699 ± 0.082/0.637 ± 0.030 in LUAD. MSFM has the higher external mean in HCC, whereas BEPH has the higher external mean in LUAD. These results suggest that explicitly encoded hand-crafted morphological features

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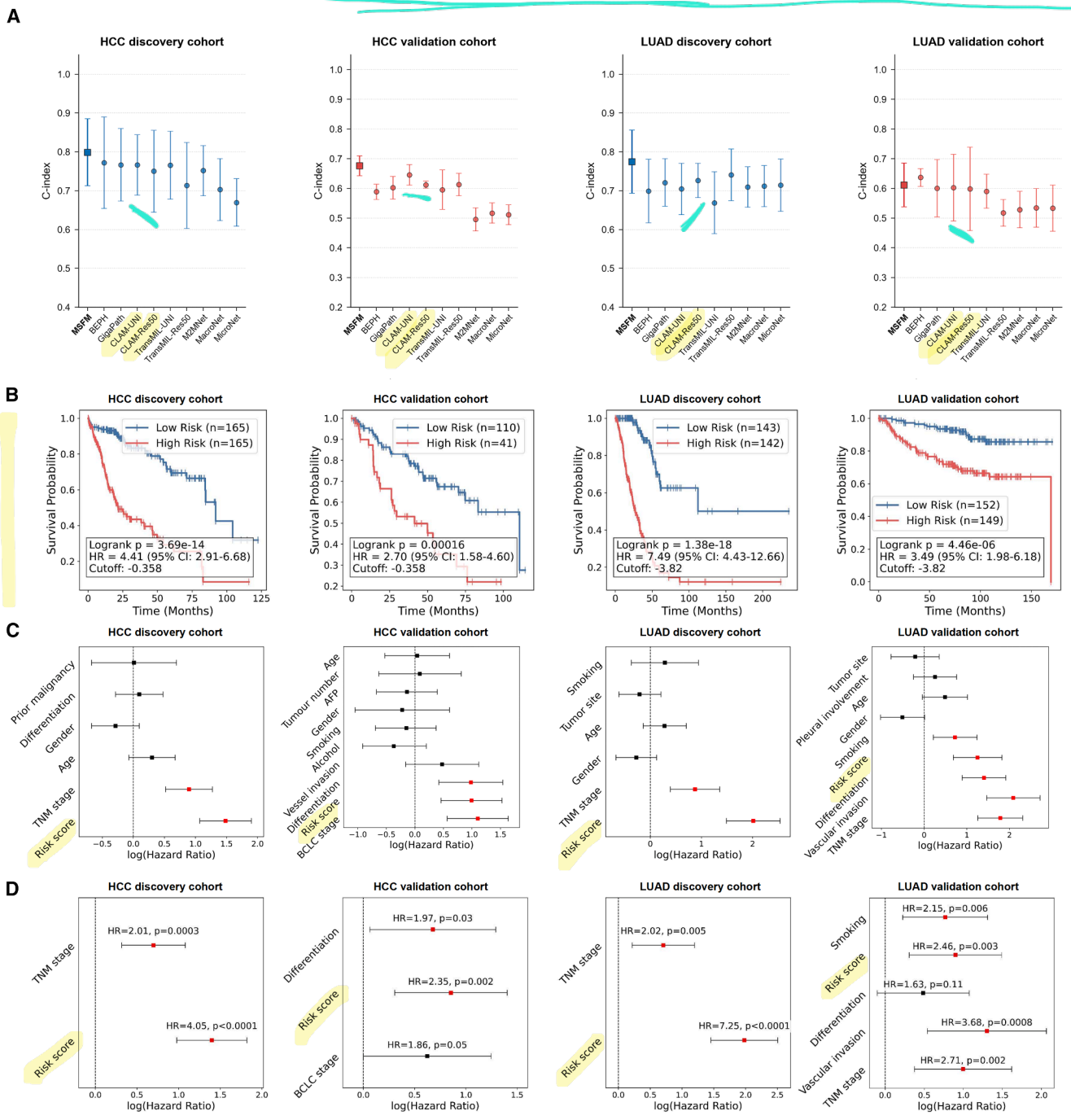

Figure 2. Performance evaluation of MSFM compared with baseline models and clinical integration analysis

(A) Point-range plots show mean C-index ± standard deviation across the ten prespecified TCGA fold-specific models for MSFM, BEPH, and the original deep learning baselines. Discovery values are internal held-out-fold metrics; external values are obtained by applying the corresponding TCGA-selected models to KMMUFH or GDPH without external model selection or tuning.

(B) Kaplan-Meier survival curves show risk stratification into high- and low-risk groups, using the discovery-cohort median MorphX risk-score cutoff transferred unchanged to the validation cohorts.

(C) Forest plots show univariable Cox regression results for the high-versus-low MorphX risk group and clinical variables (red highlights: $p < 0.05$).

(D) Forest plots show multivariable Cox regression results for the high-versus-low MorphX risk group after adjustment for cohort-specific clinical variables (red highlights: $p < 0.05$).

P values in (B) are calculated using log-rank tests. Horizontal bars in (C) and (D) indicate 95% confidence intervals.

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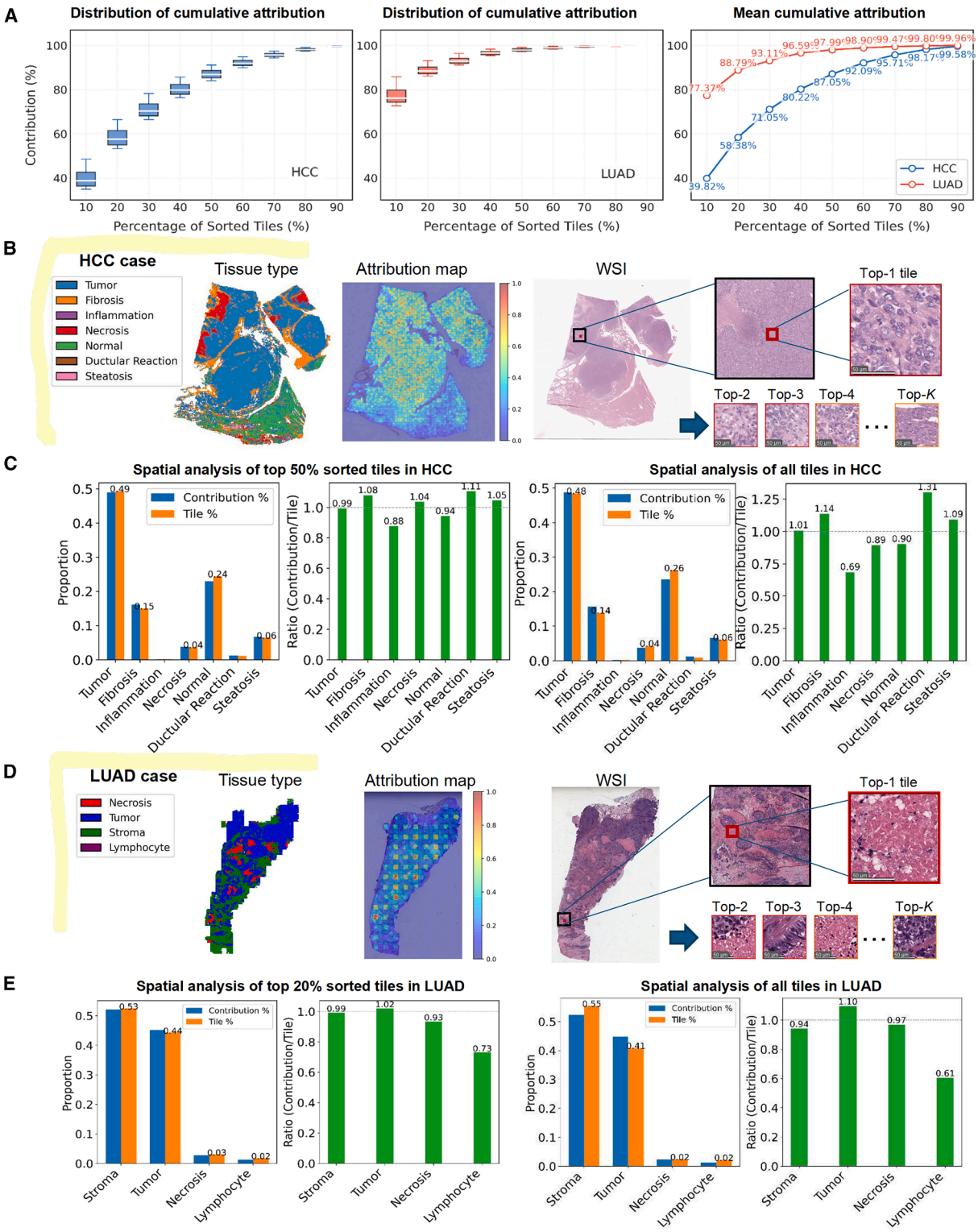

Figure 3. Spatial prognostic patterns and tissue-type-specific contribution analysis

(A) Boxplots showing the cumulative attribution percentage growth with incrementally added sorted tiles (10% intervals) in the discovery cohorts. Each observation is one processed WSI-level cumulative-attribution value (n = 330 HCC WSIs, and n = 285 LUAD WSIs; one WSI per modeled patient); boxes span the 25th–75th percentiles, center lines mark medians, and whiskers span the 5th–95th percentiles. The cohort-averaged trend line is shown at right.

(B) Visualization pipeline for identifying top-K tiles through spatial attribution mapping (HCC case).

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can provide competitive prognostic information, while retaining explicit morphology-level interpretation (Figure 2A; Table S3; Data S2).

We train a final MSFM using the full discovery cohort and refer to its output as the MorphX-derived risk score (Figures 2B–2D). The score yields external C-indices of 0.683 (95% confidence interval [CI]: 0.610–0.753) in KMMUFH and 0.670 (95% CI: 0.600–0.732) in GDPH. Kaplan-Meier analyses show significant external stratification (hazard ratios [HRs] = 2.70 in KMMUFH and 3.49 in GDPH; both log rank $p < 0.0005$; Figure 2B). After adjustment for cohort-specific clinical covariates, the high-versus-low risk group remains associated with OS (HRs = 2.35 and 2.46, respectively; Figures 2C and 2D; Table S4). Continuous-score clinical-integration analyses are shown in Figures S2J and S2K. In KMMUFH and GDPH, the clinical-only/combined C-indices are 0.520/0.691 and 0.771/0.791, respectively.

Continuous-score analyses further show that the MorphX-derived risk score remains associated with OS in all four cohorts and provides prognostic information beyond the recorded clinical covariates (Figures S2J and S2K). The prespecified OS-trained MorphX score is also associated with progression-free interval (PFI) in both discovery cohorts (HRs per SD = 1.47 in TCGA-LIHC and 2.03 in TCGA-LUAD; Figures S7A–S7D; Data S1). An exploratory 3-year descriptor screen identifies an HCC OS discovery pattern that is not confirmed after external multiple-testing correction; no descriptor passes false discovery rate (FDR) correction in the LUAD OS or either PFI screen (Figures S7E–S7J).

MorphX reveals morphology-based spatial prognostic patterns

To quantify the spatial distribution of prognostic signals, we analyze the contribution of each tile within the WSI. We sort tiles in descending order based on their absolute attribution and calculate the cumulative proportion of total attribution from the top tiles. In cohort-level analysis, the top 50% of tiles in HCC contribute a mean of 87.1% of the total attribution magnitude, whereas the top 20% of tiles in LUAD contribute a mean of 88.8% (Figure 3A). Notably, HCC appears to require more tiles to characterize prognostic information, consistent with the reported spatial heterogeneity.[26,27] These results suggest that both cancer types include a large proportion of tiles with relatively low attribution. We then visualize the top-ranked tiles based on the spatial attribution maps (Figures 3B and 3D; Figures S1D–S1F). The displayed examples were selected from regions ranked by model attribution. Representative high- and low-contribution tiles across tissue types are shown in Figure S3.

We further perform tissue-type-specific analysis and find that selected-tile patterns are broadly consistent with the all-tile analysis (Figures 3C and 3E). Tumor regions make substantial absolute contributions in both HCC (48.8%) and LUAD (44.85%). For each tissue type, the contribution-to-area ratio is the percentage of the total absolute attribution divided by the percentage of mapped tissue area; values > 1 indicate an attribution share above its proportional area. In HCC, fibrosis (15.8% contribution; ratio: 1.14) and ductular reaction (1.2%; ratio: 1.31) show attribution above their proportional areas. In LUAD, stroma is the largest absolute contributor (52.28% attribution; 55.41% area), whereas tumor contributes 44.85% of attribution over 40.87% of area, giving ratios of 0.94 and 1.10, respectively. The ratio is a descriptive area normalization, rather than a measure of the predictive efficiency or causal importance. These findings motivate the tissue-specific feature analyses below.[28,29] Representative spatial-attribution and nuclear-shape examples are shown in Figure S4.

MorphX reveals key morphological features associated with prognosis

The top 10 contributing morphological features are identified by their attribution scores across patients. In both HCC and LUAD, tissue texture features, particularly those derived from the gray-level co-occurrence matrix (GLCM),[18,30] tend to receive higher attribution than nuclear shape descriptors[31,32] (Figures 4A and 4D). The most prominent feature in HCC is glcm_MCC (maximum correlation coefficient), which captures textural complexity and long-range gray-level dependence, whereas in LUAD, the top feature is glcm_DifferenceAverage (DA), which quantifies the overall variation in pixel intensity and may reflect underlying tissue heterogeneity. Consistent with this pattern, both features not only dominate the attribution ranking but also retain statistically significant, cross-cohort prognostic value when summarized at the WSI level and evaluated using Cox regression and max-log-rank-derived cutoffs (Figures 4B and 4E). Specifically, glcm_MCC in HCC shows robust risk stratification (TCGA-LIHC: HR = 2.31; KMMUFH: HR = 2.43), and glcm_DA in LUAD likewise remains prognostic across cohorts (TCGA-LUAD: HR = 1.90; GDPH: HR = 2.11). These findings are in line with prior work linking co-occurrence-based texture measures to survival across solid tumors.[26,33,34]

In addition to the top contributing features identified within each cancer type, we also compare the importance scores of all 160 morphological features between HCC and LUAD (Figure S6A). This comparison reveals substantial differences, with a subset of features showing markedly higher importance in one cancer type than in the other (for example, glcm_MCC). Conversely, many other features exhibit comparable importance scores across both cancers, suggesting some shared prognostic morphological signals. Overall, these patterns suggest that texture-based heterogeneity may constitute a major axis of prognostic variation, with partial cancer-type-specific

(C) HCC tissue analysis: (left) dual-panel comparison of tissue area distribution versus contribution percentage for top 50% tiles; (right) corresponding analysis for all tiles.
(D) Visualization pipeline for identifying top-K tiles through spatial attribution mapping (LUAD case).
(E) LUAD tissue analysis: (left) dual-panel comparison of tissue area distribution versus contribution percentage for top 20% tiles; (right) corresponding analysis for all tiles.
Scale bars in the representative tile images in (B) and (D) are 50 μm.

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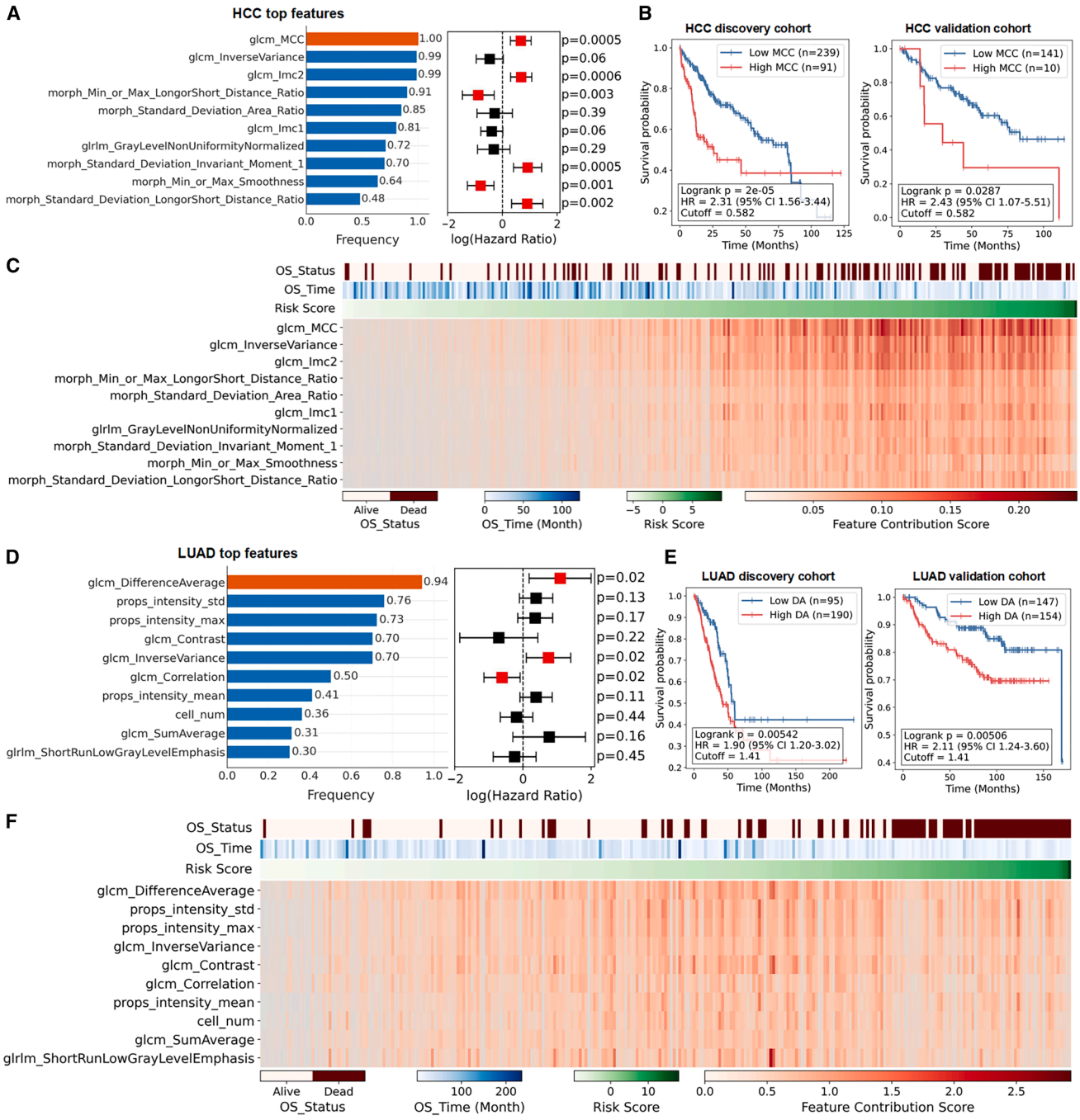

Figure 4. Prognostic value analysis of top-contributing morphological features

(A and D) Dual-panel analysis of the top 10 morphological features in the HCC and LUAD discovery cohorts: frequency distributions ranked by the attribution score, and forest plots of univariable Cox regression results with hazard ratios and 95% CIs (red highlights: $p < 0.05$).

(B and E) Kaplan-Meier survival analyses based on mean feature values across all tissue types, with cutoffs determined in TCGA by max-log-rank and transferred unchanged to external cohorts. MCC refers to the GLCM maximum correlation coefficient (glcm_MCC), and DA refers to the GLCM difference average (glcm_DifferenceAverage). P values in (B) and (E) are calculated using log-rank tests.

(C and F) Attribution trends of the top 10 features across ascending model risk scores in HCC and LUAD, showing feature-contribution patterns associated with modeled prognostic risk.

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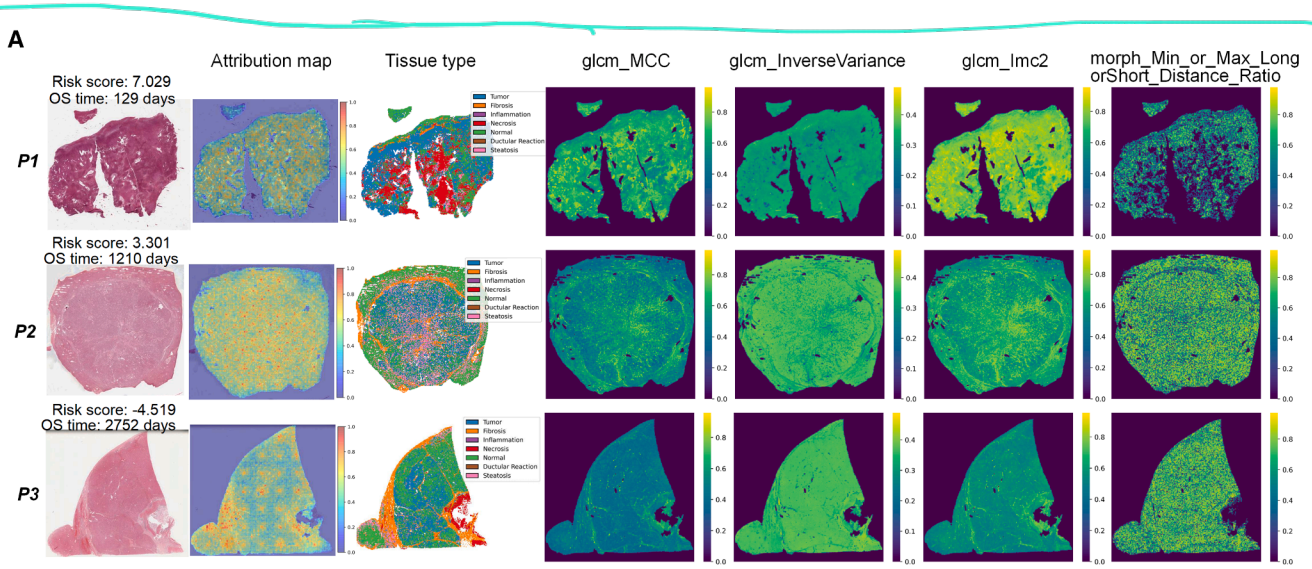

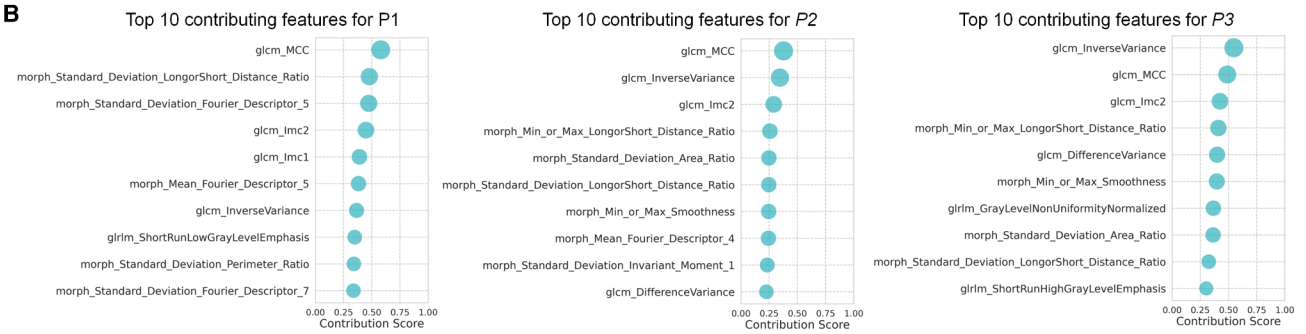

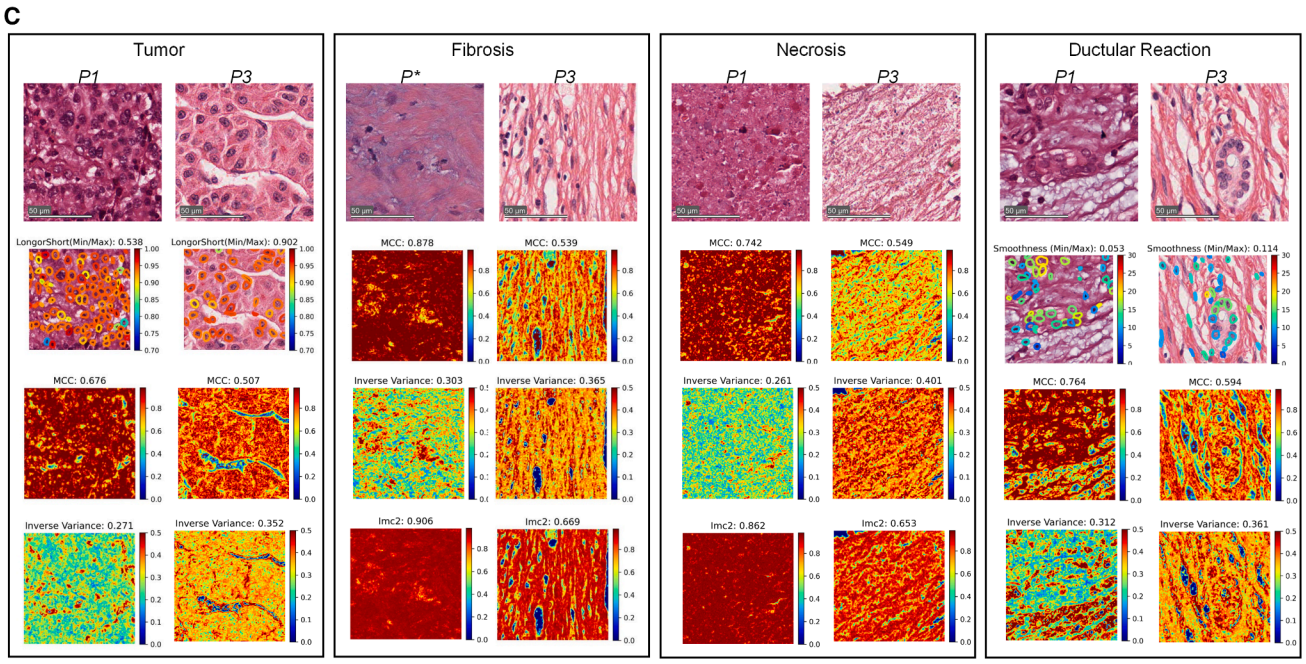

Figure 5. Distribution and visualization of prognostic morphological features in HCC

(A) Discriminative feature distributions across WSIs for three representative HCC patients with different survival outcomes illustrate differences between the high- and low-risk groups.

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emphasis, and highlight the importance of considering cancer-type-specific modeling strategies when leveraging morphological features for prognosis. The MorphX framework can be reused across cancers, but the prognostic model should be retrained or calibrated for each cancer type.

Although texture features dominate the global attribution ranking, nuclear- and cell-level phenotypes are repeatedly represented among the top contributors and show more moderate effect sizes in univariable Cox analysis (Figures 4A and 4D). In HCC, approximately half of the global top 10 features are nuclear morphology descriptors (for example, long-to-short axis distance-ratio statistics and nuclear smoothness measures), and in LUAD, several nuclear intensity and cellularity metrics (for example, intensity dispersion and cell number) also enter the top 10 list. Visual inspection of high-attribution tiles further confirms that these descriptors capture recognizable patterns of nuclear pleomorphism and contour irregularity (Figure S4A). Taken together with previous digital pathology studies showing that nuclear morphometry encodes clinically relevant and prognostic information,[35,36] these results support a complementary role for nuclear morphology alongside texture-based features.

Texture and nuclear/cell descriptors show only modest slide-level correlations, supporting their complementary contributions (Figures S5A and S5B; Data S3). Patient-level analyses further show that higher tumor glcm_MCC is associated with higher MorphX risk in both HCC cohorts, while higher tumor props_intensity_max in TCGA-LUAD and higher tumor props_area in GDPH are likewise associated with a higher risk (Figures S5C and S5D; Data S3).

We further order patients by the risk score and visualize the contribution trends of the top 10 features across the risk continuum. In HCC, the contribution of these features increases markedly with rising risk (Figure 4C). In LUAD, the contribution shows only a modest overall increase and attenuates in the highest-risk patients, indicating a broader combination of prognostic features (Figure 4F).

We additionally compute the top 10 contributing features within each tissue type to examine cross-tissue heterogeneity. The dominant features are highly consistent across tissue categories, with glcm_MCC (HCC) and glcm_DA (LUAD) repeatedly emerging as the most frequent top contributors. This suggests that texture complexity and heterogeneity represent recurrent prognostic signals, rather than being confined to a single tissue type (Figure S6B). Tissue-type-specific top-feature analyses further show that several nuclear-shape and intensity descriptors rank among the frequent contributors within tumor regions, highlighting nuclear morphology within tumor regions as an additional, tissue-grounded source of risk information.

Cross-cancer comparison identifies four shared features, 14 HCC-specific features, and 15 LUAD-specific features; the larger cancer-specific groups show higher mean external C-indices than the shared group, although the unequal group sizes preclude attributing this difference to cancer specificity alone (Figures S5F and S5H; Data S2 and S3). These restricted-feature analyses are exploratory; the primary MorphX models use the complete morphology maps containing the full prespecified feature set. Within each cancer, discovery and external feature lists also show moderate overlap (45 HCC and 30 LUAD tissue-feature entries; Figure S5G; Data S3).

Discovery and validation of potential morphology-based biomarkers

To discover potential morphology-based biomarkers, we examine morphology differences between high- and low-risk patients in both HCC and LUAD cohorts. Figures 5 and 6 summarize these differences across tissue types. In HCC, the texture-complexity metric glcm_MCC is consistently higher in high-risk patients than in low-risk patients (Figure 5A). In the representative cases shown in Figure 5A, the all-tissue mean glcm_MCC of the high-risk patient (P1) is 0.61 ± 0.15, compared with 0.44 ± 0.08 in the low-risk patient (P3). This separation is prominent at the tile level in tumor regions (0.676 vs. 0.507; $\Delta = 0.169$; Figure 5C) and remains detectable, though being weaker, in necrotic tiles (Figure 5C). Beyond tumor and necrosis, tissue-specific patterns are also present; exemplar fibrosis tiles show lower glcm_MCC values in low-risk cases (for example, 0.539 in Figure 5C), whereas high-risk cases show higher fibrosis-region texture complexity; ductular-reaction tiles in low-risk cases show higher nuclear contour smoothness (Figure 5C). In LUAD, exemplar cases show that glcm_DA is elevated in high-risk patients relative to low-risk patients (Figure 6A). For the representative cases in Figure 6A, DA in the high-risk patient (P1) is 1.77 ± 0.60, compared with 0.88 ± 0.64 in the low-risk patient (P3). The strongest contrasts localize to the necrotic and tumor regions (Figure 6C): in representative solid tumor tiles, DA differs by 0.65 between risk groups (1.91 vs. 1.26); among necrotic subtypes, tiles lacking nuclear remnants show a larger separation (2.32 vs. 1.51). Visible DA differences are also present in stromal tiles between risk groups (Figure 6C).

Guided by these qualitative observations and the tissue-type-specific attribution analysis, we next systematically screen the tissue-type-specific top features and, within each tissue type, select the first feature that meets a conservative cross-cohort criterion. In HCC, we focus on glcm_MCC in tumor and necrosis regions (TUM MCC and NEC MCC), a nuclear-shape dispersion feature in fibrosis (FIB LSR-SD), and a texture feature in steatosis (STE lmc2). In LUAD, we evaluate the glcm_DA in the necrotic, tumor, and stromal regions (NEC DA, TUM DA, and STR DA), as well as lymphocyte density (LYM DEN), which we compute from the cell_num channel in lymphocyte regions and normalize by total tissue area. The full screening rule is provided in STAR Methods.

These tissue-type-specific candidates show significant and directionally consistent stratification between high- and low-risk patients in the discovery cohorts and, under transferred discovery cutoffs, in the external cohorts, with all log rank $p < 0.05$

(B) Bubble plots summarize patient-level heterogeneity in the top 10 contributing features, with bubble size representing attribution magnitude.
(C) Tile-level differential feature visualization highlights distinct morphological signatures. Tiles are selected from representative regions of different patients and categorized by tissue type; the dense fibrous tile is sourced from an additional high-risk patient (OS time, 556 days; risk score, 6.818; OS status, dead). Scale bars in (C) represent 50 μm.

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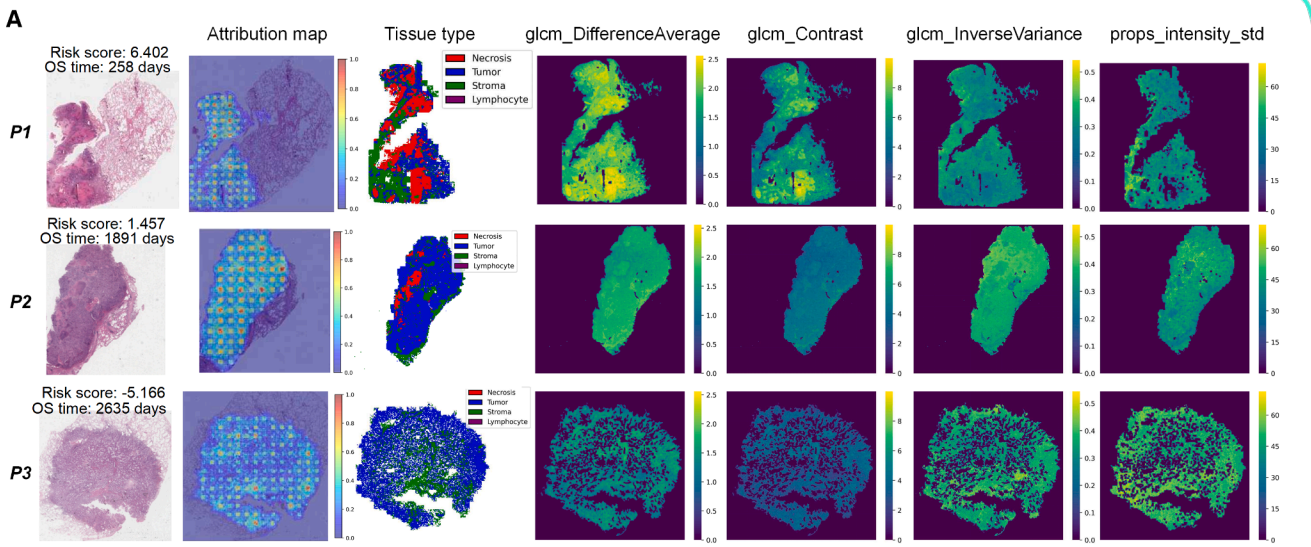

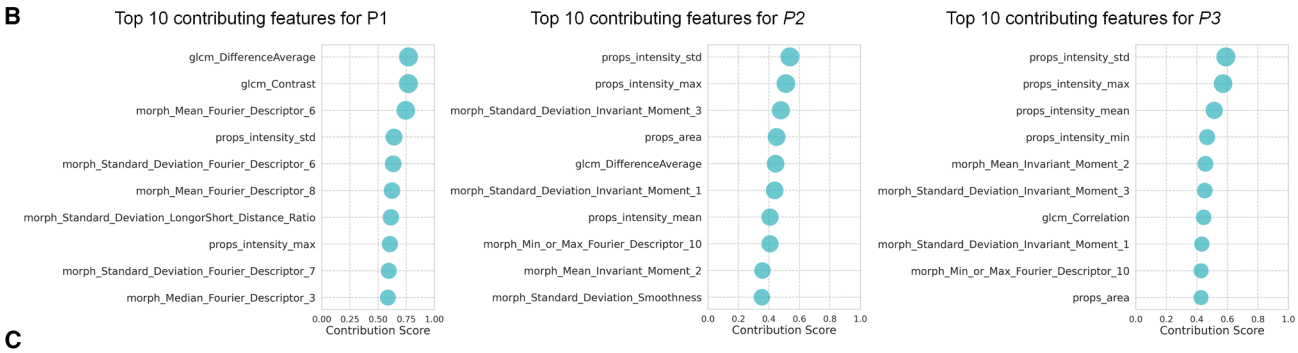

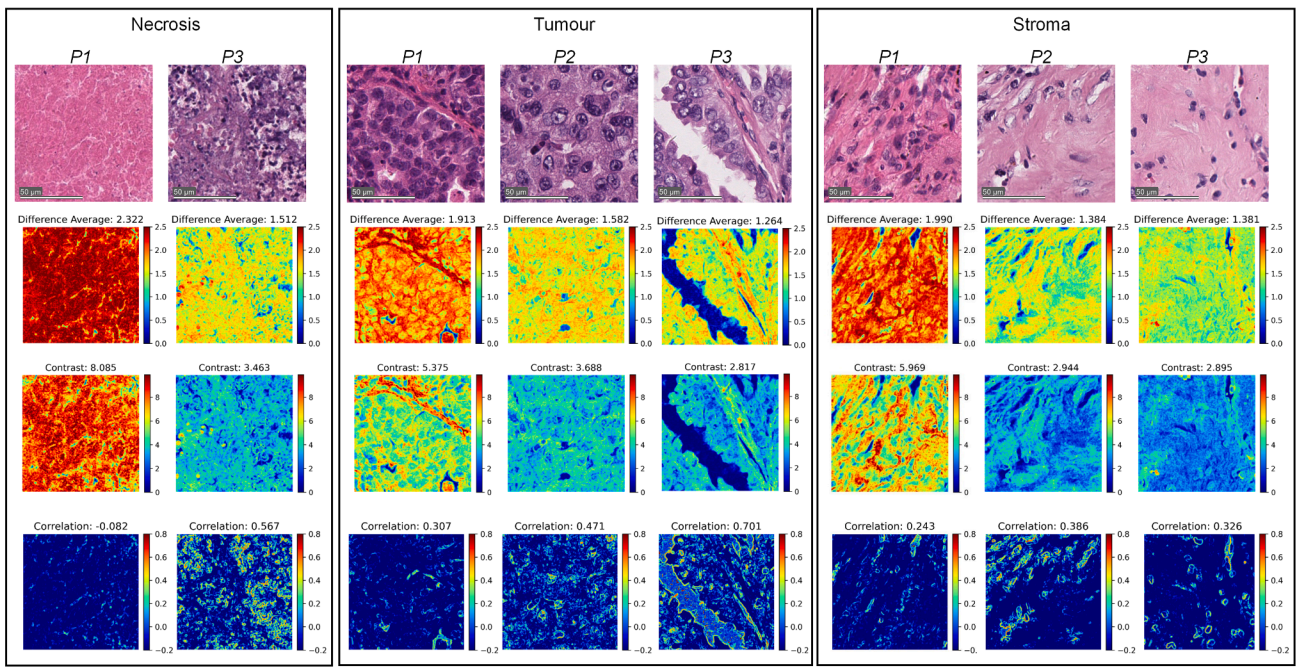

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(Figure 7). In HCC, higher TUM MCC and NEC MCC are associated with worse survival, with HRs around 2 in both discovery and validation cohorts[26,37,38]; fibrosis-related FIB LSR-SD and steatosis-related STE Imc2 provide additional tissue-type-specific risk information that remains significant after cross-cohort cutoff transfer. High-versus-low TUM MCC is associated with HRs of 2.42 and 2.15 in TCGA-LIHC and KMMUFH, respectively. The corresponding HRs are 1.79 and 2.56 for NEC MCC and 1.91 and 1.95 for FIB LSR-SD (Figures 7A–7D). In LUAD, elevated NEC DA and TUM DA, and to a lesser extent, STR DA, reflect adverse texture heterogeneity and consistently identify a high-risk group with approximately 2-fold increased hazard, whereas higher LYM DEN is associated with improved survival, consistent with a protective lymphocyte-related signal.[28,29] NEC DA yields HRs of 2.45 and 2.27 in TCGA-LUAD and GDPH, respectively. The corresponding HRs are 1.70 and 1.73 for TUM DA, 1.80 and 1.73 for STR DA, and 0.15 and 0.27 for LYM DEN (Figures 7E–7H). Taken together, these tissue-type-specific candidates span the tumor, necrotic, fibrotic, steatotic, stromal, and lymphocyte compartments and link quantitative morphology to survival across HCC and LUAD cohorts.

To screen combinations of features more systematically, we evaluated the four candidate features per cancer and identified TUM MCC + FIB LSR-SD in HCC and NEC DA + LYM DEN in LUAD. The discovery-fitted coefficients were applied unchanged to the external cohorts, where both two-feature scores remained associated with survival, although an alternative HCC pair had nearly identical support (Figures S5I–S5K; Table S5; Data S3). In multivariable Cox sensitivity analyses, the two-feature scores remained associated with survival in both discovery cohorts and KMMUFH, whereas the association in GDPH was borderline ($p = 0.0507$; Figure S5L; Table S5).

Targeted analyses in RNA-matched TCGA cases provide molecular context for overall MorphX risk and the components retained by the two-feature scores. HCC risk and score components are associated with proliferative, hypoxia, angiogenesis, and extracellular matrix programs, whereas LUAD NEC DA and LYM DEN are associated with proliferative and immune-related programs, respectively (Figures S5M–S5N; Data S3). These post-hoc associations provide cohort-scale, rather than spatial or causal, evidence.

DISCUSSION

Deep learning and pathology foundation models based on WSIs continue to emerge and offer new tools for prognosis prediction, yet few are translated into clinical practice.[39,40] A major barrier is the lack of transparency and interpretability, as the abstract deep features learned by these models often lack clear biological meaning. Some studies have attempted to improve interpretability by first using deep learning models to highlight high-attention regions and then qualitatively inspecting the tissue morphology within these regions to construct post-hoc explanations. However, these subjectively derived morphological descriptions are not incorporated into the model itself and, thus, do not fundamentally alter its "black-box" nature.[9] Other approaches enhance interpretability by utilizing tissue type maps, including histologically meaningful tissue category maps and cluster-based tissue distribution maps derived from unsupervised learning, to establish prognostic associations.[10,41] However, they still fail to elucidate the prognostic relevance of the more detailed morphological patterns, such as tissue texture and cellular morphology, resulting in limited interpretability. These limitations highlight the need for an approach that seamlessly integrates intrinsically interpretable morphological features directly into the predictive model itself.

Despite the strong performance of pathology foundation models such as GigaPath and BEPH,[21,22,25] MSFM remains competitive under the same discovery-fold and external-validation design. MSFM has a higher HCC external mean than BEPH (0.676 versus 0.589), whereas BEPH has a higher LUAD mean (0.637 versus 0.612; Figure 2A; Table S3). Unlike these latent-feature models, MorphX operates in an explicitly morphological input and attribution space. The MorphX risk score remains associated with OS after cohort-specific clinical adjustment, and the continuous-score analyses show higher apparent C-indices after adding MorphX to the recorded clinical covariates (Figure S2K). These results support the use of explicit morphological feature maps to balance prognostic performance and interpretability.

Although MorphX builds on explicitly hand-crafted morphological features, it goes beyond conventional feature selection[26] in two important aspects. First, instead of aggregating tile-level features into predefined slide-level summaries, MorphX learns a spatially resolved, non-linear mapping from morphological feature maps to patient outcome. This enables the model to capture complex interactions between features, tissue types, and spatial context that are difficult to encode manually.[18] Second, by combining this deep model with attribution-based xAI, MorphX localizes which tiles and which feature-tissue combinations drive risk predictions and uses these patterns to derive tissue-type-specific candidate biomarkers. These candidates are not simply the result of global feature-outcome association but emerge from a systematic, model-guided search over the spatial and tissue-type-specific morphological landscape.

One of the most prominent findings from our xAI-based analysis is the prognostic association of necrotic-region texture features.[37,38] Our analysis of attribution maps shows enrichment of necrotic areas within high-contribution regions (Figures 3B and 3D), in agreement with studies in which quantitative assessment of necrotic architecture, such as fractal dimension and

Figure 6. Distribution and visualization of prognostic morphological features in LUAD

(A) Discriminative feature distributions across WSIs for three representative LUAD patients with different survival outcomes illustrate differences between the high- and low-risk groups.

(B) Bubble plots summarize patient-level heterogeneity in the top 10 contributing features, with bubble size representing attribution magnitude.

(C) Tile-level differential feature visualization highlights distinct morphological signatures. Tiles are selected from representative regions of different patients and categorized by tissue type. Scale bars in (C) represent 50 μm.

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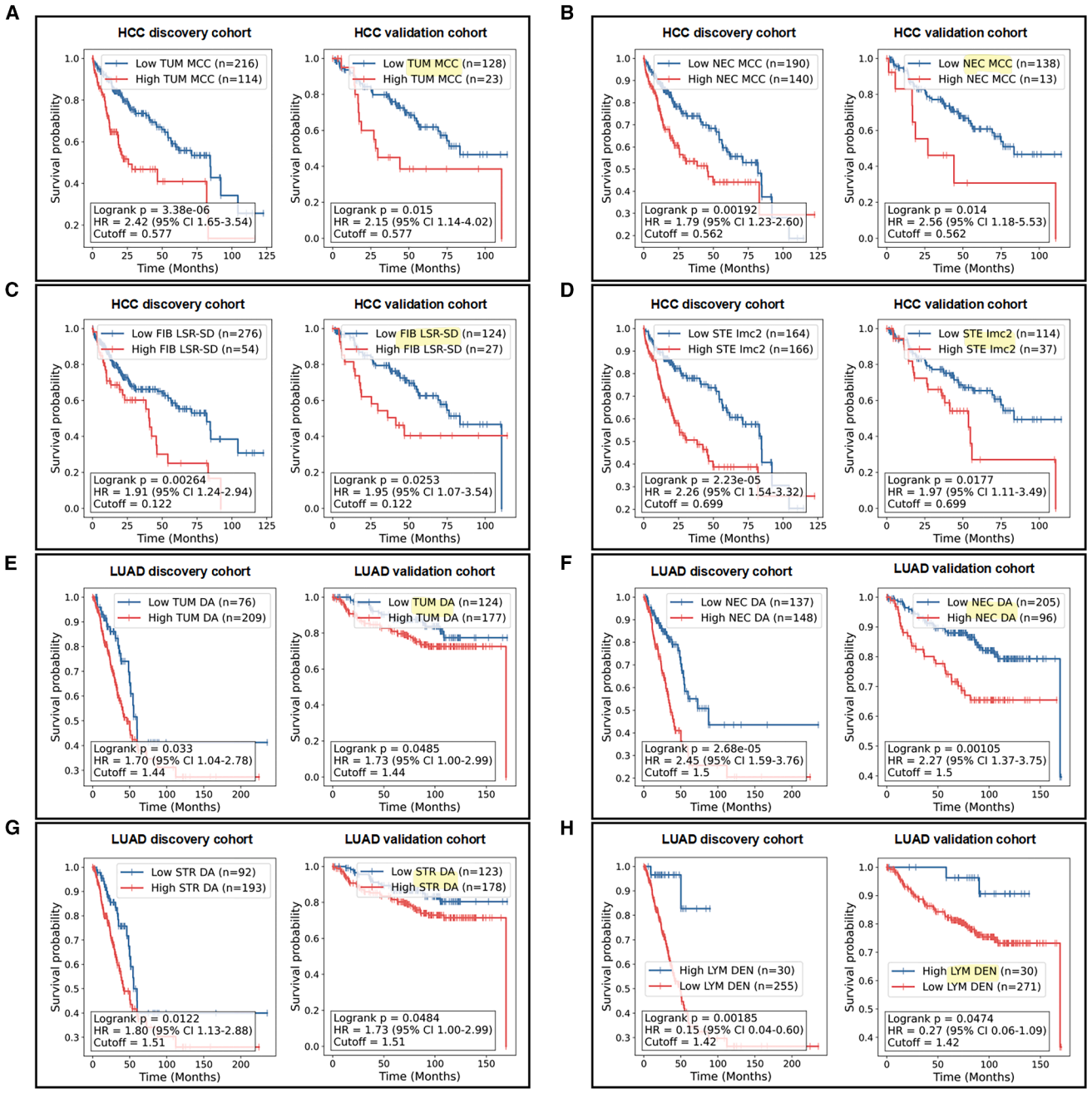

Figure 7. Cross-cohort screening of tissue-specific morphology candidates

(A–H) HCC candidates are TUM MCC, NEC MCC, FIB LSR-SD, and STE Imc2; LUAD candidates are NEC DA, TUM DA, STR DA, and LYM DEN. Shown are the tissue-specific candidate analyses; the LYM DEN result is reported in (H). Kaplan-Meier curves use discovery-derived cutoffs transferred unchanged to external cohorts. All eight candidates meet an exploratory rule requiring nominal discovery log rank $p < 0.05$, direction consistency, and nominal external log rank $p < 0.05$ after cutoff transfer. Because the external outcome contributed to candidate retention, these curves are cross-cohort screening results, rather than untouched independent validation. Higher LYM DEN is directionally protective in both cohorts, although its external Cox 95% CI is wide and includes 1.
The restricted two-feature score analysis is shown in Figure S5 and Data S3.

lacunarity on imaging, is linked to survival and motivates further investigation into their morphological characteristics.[34] Visualization of the feature heatmaps shows that necrotic regions exhibit higher texture-feature values in representative high-risk patients (Figures 5A and 6A). We then compare texture features of necrotic areas between high- and low-risk patients and quantify structural disorganization and grayscale heterogeneity, using explicit hand-crafted features (Figures 5C and 6C). These

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differences are directly interpretable and spatially traceable, which are hallmarks of hand-crafted features.

Our approach may capture complementary information compared with existing necrosis-related biomarkers such as necrosis extent (NEC) and tumor-necrosis distance (TND).[11,37] While NEC and TND primarily describe the spatial extent of necrosis and its proximity to tumor regions, our texture-based candidates place greater emphasis on the internal textural complexity of necrotic regions, which may provide finer-grained insights into the necrosis dynamics and its interplay with tumor biology (Figures 7B and 7F). The texture features may partly reflect different phases of necrosis and the degree of structural disruption before overt necrotic disintegration. Taken together, these necrosis-related texture candidates complement existing measures and may help characterize tumor malignancy more comprehensively.[32,42]

Building on this discovery, the identification of necrosis-related texture candidates further motivates a broader search for prognostic patterns in other tissue types. By extracting high-contribution regions within specific categories, we identify additional candidate features across tumor, fibrosis, stroma, and steatosis (Figure 7). The exploratory HCC score combines tumor texture with fibrosis-region nuclear-shape variability, but an eligible FIB LSR-SD + STE Imc2 alternative has nearly identical discovery Akaike information criterion (AIC); so, the present data do not establish a unique HCC pair. The LUAD score combines adverse necrotic texture with a negatively weighted lymphocyte-density component. Their distinct compartments provide a plausible rationale for complementary information, and the targeted bulk-RNA results offer molecular context.[43]

Beyond cohort-level analysis, our framework enables patient-level interpretation. For each case, we identify both the most influential morphological features (Figures 5B and 6B) and the specific spatial regions in the WSI that drive the prediction (Figures 3B and 3D). This dual perspective allows the generation of individualized morphological profiles. These displays provide case-specific evidence for future blinded pathologist-reader studies.

In conclusion, our study demonstrates that integrating explicitly encoded morphological characteristics with deep learning through MorphX achieves a favorable balance between prognostic performance and interpretability. By fusing spatially aligned, hand-crafted morphological feature maps with deep learning, we provide a concrete example of how explicit WSI-level morphological maps can be combined with deep models to build high-performance yet interpretable AI systems in computational pathology.

Limitations of the study

This retrospective study includes one external cohort per cancer and does not establish prospective clinical utility. Comparable detailed treatment regimens are not consistently available across the four cohorts, limiting treatment-adjusted interpretation. The tissue-specific candidates and two-feature scores remain exploratory because external outcomes contribute to candidate retention. The matched bulk-RNA analysis provides cohort-level molecular context, rather than spatial or causal validation, and organ-specific tissue taxonomies and tile-level aggregation of nuclei limit cross-cancer and lineage-specific interpretations.

These limitations motivate several directions for future research. The current set of hand-crafted features, although biologically grounded, represents a predefined morphological vocabulary. Future work could extend MorphX to identify subtle morphological phenotypes that are not captured by conventional pathological descriptions. Clinical translation will also require intuitive interfaces that enable pathologists to examine model predictions and their contributing features. Finally, the generalizability and clinical value of MorphX and the identified candidate features should be assessed in large-scale prospective studies across diverse populations and cancer types.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Cheng Lu (lucheng@gdph.org.cn).

Materials availability

This study does not generate new physical reagents, cell lines, animal models, or other tangible biological materials.

Data and code availability

- H&E-stained WSIs and clinical data from TCGA-LIHC and TCGA-LUAD are available through the Genomic Data Commons (https://portal.gdc.cancer.gov/). H&E-stained WSIs and clinical data from KMMUFH and GDPH contain institutional patient records and are available through controlled institutional access. Researchers may request access by emailing the lead contact, specifying the requested data and intended research use. Access is subject to applicable institutional policies, ethics approval, and data-use approval.
- The MorphX code repository is publicly available at https://github.com/FearlessII/MorphX.
- Any additional information required to reanalyze the data reported in this work paper is available from the lead contact upon request.
- Supplemental data are provided as three separate Excel files: Data_S1.xlsx (Data S1), Data_S2.xlsx (Data S2), and Data_S3.xlsx (Data S3). Supplemental information includes Figures S1–S7 and Tables S1–S5. Datasets, software, segmentation models, and code resources are listed in the key resources table.

ACKNOWLEDGMENTS

The authors would like to acknowledge the support provided by MediAI Hub, an advanced medical image analysis software developed and maintained by MediaLab. This work was funded by the National Natural Science Foundation of China (no. 82272084 to C.L.); National key research and development program intergovernmental key projects (no. 2023YFE0114300 to C.L.); National Key R&D Program of China (no. 2023YFC3402800 to C.L.); and National Science Foundation for Young Scientists of China (no. 82502463 to H. Lin).

AUTHOR CONTRIBUTIONS

Conceptualization, Zhishun Liu and Q.W.; methodology, Zhishun Liu, Q.W., D.Y., X.L., Y.W., and Y.Z.; software, Zhishun Liu, Q.W., and Y.Z.; validation, Zhishun Liu, Q.W., and X.L.; formal analysis, Zhishun Liu, Q.W., and X.L.; investigation, Zhishun Liu, Q.W., H. Lin, C.T., and H. Liu; resources, H. Lin and H. Liu; data curation, Zhishun Liu, Q.W., H. Lin, and C.T.; visualization, Zhishun Liu, Q.W., J.C., and C.T.; supervision, Z. Li, Zaiyi Liu, A.M., and C.L.; writing – original draft, Zhishun Liu; writing – review & editing, Zhishun Liu, Q.W., H. Lin, D.Y., C.T., X.L., Y.Z., J.C., Y.W., Z.C., H. Liu, Z. Li, Zaiyi Liu, A.M., and C.L.

DECLARATION OF INTERESTS

A.M. is a Research Career Scientist at the Atlanta Veterans Affairs Medical Center and an equity holder in Picture Health, Elucid Bioimaging, and Inspirata Inc. He serves on the advisory board of Picture Health, consults for Takeda Inc., and has sponsored research agreements with AstraZeneca and Bristol Myers-Squibb. His technology has been licensed to Picture Health and Elucid Bioimaging. He is also involved in one R01 grant with Inspirata Inc.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this manuscript, the authors used OpenAI ChatGPT solely for English-language editing and polishing. The scientific content, analyses, interpretations, and conclusions were written by the authors. After using this tool, the authors reviewed and edited the text as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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  - Morphological feature map generation
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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.xcrm.2026.103126.

Received: April 13, 2026
Revised: August 14, 2026
Accepted: October 4, 2026

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
TCGA-LIHC H&E WSIs and clinical data	TCGA/GDC	RRID:SCR_014514; https://portal.gdc.cancer.gov/
TCGA-LUAD H&E WSIs and clinical data	TCGA/GDC	RRID:SCR_014514; https://portal.gdc.cancer.gov/
KMMUFH HCC cohort	The First Affiliated Hospital of Kunming Medical University	N/A
GDPH LUAD cohort	Guangdong Provincial People's Hospital	N/A
TCGA bulk RNA-seq STAR-count files	NCI Genomic Data Commons	RRID:SCR_014514; https://portal.gdc.cancer.gov/
MSigDB Hallmark, GO BP, and KEGG legacy gene sets	MSigDB	RRID:SCR_016863; https://www.gsea-msigdb.org/gsea/msigdb/
Software and algorithms		
MorphX code	This paper	v0.1.0; DOI: https://doi.org/10.5281/zenodo.22857166
HoVer-Net nucleus segmentation model	Graham et al.[44]	https://doi.org/10.1016/j.media.2019.101563
PaSegNet HCC tissue segmentation model	Liang et al.[11]	https://doi.org/10.1038/s42256-023-00635-3
Multi-Layer Pseudo-Supervision LUAD tissue segmentation model	Han et al.[45]	https://doi.org/10.1016/j.media.2022.102487
HistoQC WSI quality-control software	Janowczyk et al.[46]	RRID:SCR_025780; https://doi.org/10.1200/CCI.18.00157
Python	Python Software Foundation	v3.10.9; RRID:SCR_008394; https://www.python.org/
PyTorch	PyTorch	v1.12.1; RRID:SCR_018536; https://pytorch.org/
pandas	pandas	v1.5.3; RRID:SCR_018214; https://pandas.pydata.org/
statsmodels	statsmodels	v0.13.5; https://www.statsmodels.org/
NumPy	NumPy	v1.23.5; RRID:SCR_008633; https://numpy.org/
openpyxl	openpyxl	v3.0.10; https://openpyxl.readthedocs.io/
Matplotlib	Matplotlib	v3.7.0; RRID:SCR_008624; https://matplotlib.org/
SciPy	SciPy	v1.10.0; RRID:SCR_008058; https://scipy.org/
scikit-learn	scikit-learn	v1.2.1; RRID:SCR_002577; https://scikit-learn.org/
Pillow	Pillow	v9.4.0; RRID:SCR_023337; https://python-pillow.org/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Study design and cohorts

This study is a retrospective, observational analysis of de-identified human H&E-stained WSIs and matched clinical outcomes. No randomized clinical trial or prospective intervention is performed. Sample size is determined by the availability of eligible WSIs and

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complete survival outcomes; completeness of downstream clinical covariates is analysis-specific and does not determine eligibility for image modeling. No pre-study power calculation is performed.

The study analyzes 1,067 patients with hepatocellular carcinoma (HCC) or lung adenocarcinoma (LUAD). HCC analyses include 330 patients from TCGA-LIHC as the discovery cohort and 151 patients from KMMUFH as an independent external validation cohort. LUAD analyses include 285 patients from TCGA-LUAD as the discovery cohort and 301 patients from GDPH as an independent external validation cohort. All WSIs are H&E-stained slides from primary tumor resection specimens.

For the TCGA cohorts, slides and associated clinical data are obtained through TCGA/GDC. TCGA samples are collected by participating institutions under their local human-subjects protocols and informed consent procedures. External cohorts are collected under Institutional Review Board approval at KMMUFH (IRB KYLX2024-171) and GDPH (IRB KY-Z-2022–241–03). Participants in the external cohorts provide written informed consent before enrollment. All analyses are performed on de-identified data.

Eligible cases for analyses are required to have (i) an H&E WSI from a primary surgical resection with sufficient analyzable tissue following quality control and (ii) complete OS time and censoring status. Cases are excluded before analysis if they lack OS data, fail WSI quality control because of severe artifacts or insufficient tissue, or lack required linkage between WSI and clinical records. Clinical covariates are used only for downstream clinical-integration analyses. Cohort characteristics and non-missing denominators are reported in Table S1; selected clinical-field availability is visualized in Figure S2B; Data S1 provides the cohort-characteristics, model-variables, treatment-availability, and field-availability worksheets.

Clinical variables and outcomes

The primary outcome is OS, defined as time from primary surgery to death from any cause or last follow-up. Event status is coded as 1 for death and 0 for censoring. OS time is analyzed in months for all four cohorts using the available clinical fields.

The score parameterization is analysis-specific. Figures 2B–2D and Table S4 use the high-versus-low MorphX risk group in survival-group and Cox analyses, whereas the analyses in Figures S2J and S2K use the continuous final MorphX score for per-SD Cox and C-index-based clinical-integration analyses. The primary high-versus-low multivariable Cox models use the following clinical covariates: grouped TNM stage (I–II vs. III–IV) in TCGA-LIHC; grouped BCLC stage (0–1 vs. 2–4) and differentiation (G1/G2 vs. G3/G4) in KMMUFH; grouped TNM stage in TCGA-LUAD; and grouped TNM stage, vascular invasion, differentiation, and smoking in GDPH (Table S4; Data S1). The continuous-score analyses use selected cohort-specific clinical covariates with predefined categorical coding: grouped TNM stage in TCGA-LIHC, grouped BCLC stage in KMMUFH, grouped TNM stage in TCGA-LUAD, and grouped TNM stage, vascular invasion, differentiation, and smoking in GDPH. The scores are z-transformed within cohort so that HRs are reported per 1-SD increase; this transformation does not change patient ordering, refit the score, or select a cutpoint. In both the primary high-versus-low and supplementary continuous-score clinical models, all missing included clinical covariates are handled by MICE (m = 20). The analysis populations are 330 patients/112 OS events in TCGA-LIHC, 151/56 in KMMUFH, 285/98 in TCGA-LUAD, and 301/62 in GDPH. The KMMUFH continuous-score model additionally uses Firth bias-reduced Cox because ordinary Cox remains affected by complete separation in some completed datasets (Figure S2J). Stage variables are entered as grouped categorical terms rather than one-step linear variables.

Gender

Gender is evaluated as an available clinical covariate, and its cohort distribution is reported in Table S1.

Randomization, blinding, and bias reduction

This is a retrospective study with non-randomized enrollment. For model development, patients in each discovery cohort are randomly assigned once to a single prespecified 10-fold cross-validation partition using a fixed random seed. The same patient-to-fold assignments are reused across MorphX and eligible baseline comparisons. KMMUFH and GDPH serve as external-validation cohorts and are excluded from model adaptation, checkpoint selection, and hyperparameter selection.

WSIs and clinical variables are de-identified and linked by study IDs. WSI quality control, tiling, tissue segmentation, and hand-crafted feature extraction are performed with automated pipelines independent of survival outcomes. Outcome information is used only during model training and statistical evaluation. For the primary MorphX model, comparator, and risk-score analyses, checkpoints, hyperparameters, and cutoffs are defined entirely in the discovery cohorts. The tissue-specific candidate-retention rule is a separate explicitly cross-cohort screen described below and is not treated as untouched independent validation.

METHOD DETAILS

WSI preprocessing and tissue quality control

Published preprocessing components include PaSegNet for HCC tissue maps,[11] HoVer-Net for nucleus segmentation,[44] the Multi-Layer Pseudo-Supervision model for LUAD tissue maps,[45] and HistoQC for WSI quality control.[46] WSIs are processed at 10× magnification after HistoQC screening and automated analyzable-tissue masking. Feature-map construction then follows parallel tissue-map and hand-crafted-descriptor branches. In the hand-crafted branch, overlapping H&E tiles are sampled with a tile size of k0 = 75 pixels and a step size of s = 37 pixels. The texture subbranch computes 40 gray-Level co-occurrence matrix (GLCM) and gray-level run length matrix (GLRLM) descriptors directly from H&E pixels, and the nuclear/cell subbranch uses HoVer-Net only to obtain

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nucleus masks before the study's hand-crafted feature-extraction code computes 120 tile-level aggregate nuclear/cell descriptors from all segmented nuclei in each tile. In parallel, HCC tissue type maps are generated with PaSegNet[11] and contain seven tissue types (C = 7). For LUAD, the published Multi-Layer Pseudo-Supervision model[45] defines four named tissue classes—tumor, stroma, necrosis, and lymphocyte tissue—and no fifth normal class. The stored LUAD representation additionally contains an auxiliary shared-code channel that marks both derived normal and blank/background positions. The analyzable-tissue mask disambiguates this shared code spatially: within analyzable tissue, positions outside the union of the four supervised tissue masks form the derived normal region; outside analyzable tissue, the same code denotes blank/background. This encoding convention does not create a fifth supervised tissue class. MorphX nevertheless retains all analyzable tissue in the complete input, including the derived normal region (Figure S1E). Here, "lymphocyte tissue" is a tissue-class label for regions dominated by lymphocytic tissue; it is not an individual-cell, graded stroma-with-lymphocytes, or tertiary-lymphoid-structure classifier, and stroma remains a separate tissue class. In the LUAD derived-normal-region-only analysis, all 165 channels are retained only within this derived normal region and set to zero elsewhere, as detailed below. The stored HCC arrays additionally contain an empty/background channel. The 160-channel hand-crafted map is assembled at the original tile coordinates, resized/aligned to the tissue-map resolution, and concatenated with the tissue-related channels; the complete construction is shown in Figures S1D and S1E. The published organ-specific tissue models are used because their liver and lung label sets represent different histological appearances and semantic tissue vocabularies. Once the parallel branches are spatially aligned and concatenated, the same downstream MFM/MSFM and attribution workflow is applied to both cancers.

Hand-crafted feature extraction operates at 10× magnification using overlapping 75-pixel tiles with a 37-pixel step. The feature-map arrays contain 168 HCC channels and 165 LUAD channels: 40 tile-level texture descriptors, 120 tile-level aggregate nuclear/cell descriptors, and 8 HCC or 5 LUAD stored tissue-related channels. HCC tissue-related channels comprise seven tissue types plus empty/background; LUAD channels comprise four named tissue classes plus an auxiliary shared-code channel for derived normal and blank/background positions. The 40 texture descriptors comprise 24 GLCM and 16 GLRLM features computed directly from H&E tile pixels. HoVer-Net supplies nucleus masks; the study's feature-extraction code then computes the 120 nuclear/cell descriptors from all segmented nuclei within each tile. These comprise 100 aggregated nuclear-morphology summary channels (mean, median, one prespecified minimum-or-maximum summary, and standard deviation for each of 25 measurements), 18 region-property descriptors spanning nuclear geometry, centroid, and intensity, and two cellularity measures (cell count and cell-area ratio). Tissue-type maps provide tile-level compartment context. No tumor, immune, or stromal lineage classifier is applied at the single-nucleus level.

For the two cancer-specific ablation analyses assessing the role of normal tissue, spatial masking is applied within the same retraining and evaluation framework. For LUAD, the published four-class tissue model has no separate supervised normal class. The derived normal-region mask comprises positions inside the analyzable-tissue mask and outside the union of the four supervised tissue masks (tumor, stroma, necrosis, and lymphocyte tissue). This intersection excludes blank/background positions that carry the same auxiliary code. All 165 channels are retained only within the derived normal region and set to zero elsewhere before MFM retraining with the same prespecified TCGA-LUAD patient-to-fold assignments. GDPH is evaluated only after TCGA checkpoint selection and without external tuning. For HCC, all 168 channels at mapped normal tiles are set to zero while the other tissue compartments are retained. Each fold-specific model is evaluated in the same 151 KMMUFH WSIs without external tuning. External C-indices are summarized as mean ± sample SD across ten fold-specific models.

The 160-channel hand-crafted vocabulary was prespecified before survival modeling from established, interpretable histopathological descriptor families[18,30,32]: gray-level co-occurrence and run-length texture, nuclear morphometry, region properties, and cellularity. These descriptors capture tissue architecture, local heterogeneity, nuclear form, intensity, and cell content and can be mapped back to histological phenotypes; they were not selected using survival associations or external-cohort performance. HoVer-Net supplies nucleus segmentation masks, not hand-crafted descriptor values; the study's feature-extraction code calculates the downstream tile-level aggregate nuclear/cell descriptors. After spatial alignment, tissue-type, texture, and nuclear/cell channels share the same grid, and the nuclear/cell descriptors aggregate all segmented nuclei within each tile rather than assigning tumor-cell, immune-cell, or stromal-cell lineages.

Morphological feature map generation

Tile-level feature extraction

Hand-crafted features are systematically extracted from tissue regions of the WSI using a sliding window approach. Each tile $T_{i,j}$ is defined with a fixed size $T=(k_0,k_0)$ pixels and a step size $s = \left\lfloor \frac{k_0}{2} \right\rfloor$, where tile coordinates are computed as

$$i = \left\lfloor \frac{x}{s} \right\rfloor, j = \left\lfloor \frac{y}{s} \right\rfloor, (x,y) \in \text{WSI coordinates.}$$

From each tile, we extract two complementary classes of quantitative morphological descriptors: 40 texture descriptors and 120 nuclear/cell descriptors.

Texture features are computed using both GLCM and GLRLM methods. These classical second-order statistical descriptors have long been used in histopathology to capture spatial co-occurrence and run-length patterns of gray levels, providing interpretable measures of tissue heterogeneity and architectural disorder.[18,30] The GLCM quantifies texture by measuring the frequency of

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pixel-intensity co-occurrence at a given offset, yielding statistics such as contrast, correlation, energy, and homogeneity. Similarly, the GLRLM captures the distribution of contiguous pixel runs with identical gray levels. These texture metrics are concatenated into a texture feature vector

$$f_{i,j}^{\text{tex}} \in \mathbb{R}^{d_1}.$$

Concurrently, the 120 nuclear/cell descriptor channels are computed from the segmented nucleus masks. They comprise 100 summary channels—mean, median, one prespecified minimum-or-maximum summary, and standard deviation for each of 25 nuclear-morphology measurements—18 region-property descriptors spanning geometry, centroid, and intensity, and two cellularity measures (cell count and cell-area ratio). These channels are aggregated into a nuclear/cell descriptor vector

$$f_{i,j}^{\text{nuc}} \in \mathbb{R}^{d_2}.$$

The final hand-crafted feature vector for each tile is

$$f_{i,j} = \left[f_{i,j}^{\text{tex}}, f_{i,j}^{\text{nuc}}\right] \in \mathbb{R}^{D}, D = d_1 + d_2,$$

where $f_{i,j}$ encodes both texture and nuclear/cell information for tile $T_{i,j}$.

Morphological feature map construction and fusion

For the entire WSI, hand-crafted feature maps $\mathcal{M}_{\text{hcf}}$ are constructed by preserving spatial correspondence between tiles and their original coordinates:

$$\mathcal{M}_{\text{hcf}} \in \mathbb{R}^{w_0 \times h_0 \times D}, w_0 = \frac{W}{s}, h_0 = \frac{H}{s},$$

where W and H denote the WSI dimensions. Tissue type maps $\mathcal{M}_{\text{tt}}$ provide semantic tissue characterization at a resolution of $w \times h \times C$ (for C tissue classes). To integrate these with the hand-crafted features, $\mathcal{M}_{\text{hcf}}$ is resized to (w,h,D) and concatenated channel-wise with $\mathcal{M}_{\text{tt}}$, yielding the unified morphological feature maps

$$\mathcal{M}_{\text{mf}} = \text{concat}(\mathcal{M}_{\text{tt}}, \mathcal{M}_{\text{hcf}}) \in \mathbb{R}^{w \times h \times (C+D)},$$

where $\mathcal{M}_{\text{mf}}$ denotes the final WSI-level morphological feature map used as input to the deep learning models and concat$(\cdot,\cdot)$ denotes channel-wise concatenation.

Deep learning architecture

MFM architecture

MFM processes the morphological feature maps $\mathcal{M}_{\text{mf}}$ through a CNN architecture that learns spatial associations across all feature channels while preserving local morphological patterns. The model first applies batch normalization (BN) to standardize $\mathcal{M}_{\text{mf}}$. To reduce the spatial resolution of the high-dimensional input while preserving local morphological patterns, we employ an initial 3×3 convolutional layer with stride 2 and match its input channels to $\mathcal{M}_{\text{mf}}$. Subsequent layers maintain a balance between local granularity and hierarchical feature integration.

The core feature extraction uses a residual architecture with four residual blocks, limiting parametric complexity while ensuring sufficient representational capacity.[47,48] This hierarchy produces a compact WSI-level embedding vector

$$z_{\text{M}} = f_{\text{enc}}(\mathcal{M}_{\text{mf}}),$$

where $f_{\text{enc}}(\cdot)$ denotes the encoder mapping of MFM and z_{M} is the resulting global morphological representation.

The MFM prediction module processes z_{M} through two fully connected layers: the first with BN and ReLU activation, and the second outputting a scalar logit $\tilde{r} \in \mathbb{R}$. We then apply a sigmoid function to obtain a bounded risk score

$$r = \sigma(\tilde{r}) \in (0,1),$$

where higher r indicates greater predicted hazard. For ablation, we also evaluate a variant MFM-σ in which the sigmoid is removed and the model directly outputs the unbounded logit $\tilde{r}$; this variant is used only in the ablation study to assess the effect of output activation on optimization stability and generalizability.

Survival loss

For survival prediction, we adopt the Cox proportional hazards loss, which optimizes the partial likelihood of the Cox model.[49] This loss function directly models the hazard ratio while handling right-censored data and is widely used in deep survival frameworks. For patient i with survival time t_i and censoring indicator δ_i (1 for observed events, 0 for censored), the loss evaluates the predicted risk score r_i against the risk set $R(t_i) = \{j | t_j \geq t_i\}$:

$$\mathcal{L}_{\text{cox}} = -\sum_{i:\delta_i=1}\left(r_i - \log\sum_{j \in R(t_i)} \exp(r_j)\right),$$

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where the summation is restricted to uncensored patients ($\delta_i = 1$). Here r_i denotes the risk score derived from the network output. The model defines the scalar output $\tilde{r}_i$ and the optional bounded mapping $r_i = \sigma(\tilde{r}_i) \in (0,1)$. The ablation label MFM-σ denotes omission of the sigmoid so that $r_i = \tilde{r}_i$ remains unbounded. The reported continuous-risk analyses use the unbounded scalar output as a hazard-aligned prognostic score rather than a probability.

Integrated gradients and top-tile selection

Integrated gradients attribution

To enable interpretable feature analysis in the trained MFM, we employ integrated gradients attribution[20] to quantify the contribution of each element in the input morphological feature map to the model's risk prediction. Attribution is evaluated on the discrete feature-map grid constructed with overlapping sliding windows (tile size 75 pixels; step size 37 pixels). Adjacent grid locations therefore share tissue content, and local CNN aggregation may reinforce similar responses across neighboring locations. For notational convenience, let

$$\mathbf{X} := \mathcal{M}_{\mathrm{mf}}, \mathbf{X}^{(0)}$$

denote the input morphological feature map and a baseline map of identical shape, respectively, where $\mathbf{X}^{(0)}$ is a null representation devoid of morphological signal (e.g., all zeros). The attribution is computed with respect to the predicted risk score $r(\mathbf{X})$.

For each element q in $\mathbf{X}$, its integrated gradients attribution a_q is computed as

$$a_q = \left(X_q - X_q^{(0)}\right) \cdot \frac{1}{m} \sum_{k=1}^{m} \frac{\partial r\left(\mathbf{X}^{(0)} + k/m\left(\mathbf{X} - \mathbf{X}^{(0)}\right)\right)}{\partial X_q},$$

where m is the number of integration steps (we use $m = 50$ in all experiments) and $\mathbf{X}^{(0)}$ is a zero-valued baseline map of identical shape.

The 3D attribution tensor A is then aggregated across its channel dimension to yield a 2D spatial contribution map

$$S = \sum_{c=1}^{C+D} |A^{(c)}|,$$

where $A^{(c)}$ denotes the c-th channel of A and the absolute value is taken element-wise. Thus, the spatial map ranks attribution magnitude irrespective of sign; it does not distinguish risk-increasing from risk-decreasing contributions. The map $S \in \mathbb{R}^{w \times h}$ highlights regions with large prognostic influence in the WSI.

Top-K tile selection

Let $s_{i,j}$ denote the value of S at tile index (i,j). The top-K tiles $\{T^{(k)}\}_{k=1}^{K}$ are selected as those whose indices correspond to the K largest entries of $\{s_{i,j}\}$. Each selected tile $T^{(k)}$ is then mapped back to physical WSI coordinates (x_k, y_k), and an expanded region centered at (x_k, y_k) is extracted to include adjacent tissue context. The selected tile and a larger context region centered on it are mapped back to the WSI for visualization.

Multi-scale features model

Macro-scale branch

The proposed MSFM hierarchically integrates macro-scale morphological patterns from WSI-level feature maps with micro-scale patterns from the key tiles identified above. Macro-scale feature extraction reuses the pretrained MFM encoder $f_{\mathrm{enc}}(\cdot)$ to generate the global feature vector

$$z_{\mathrm{M}} = f_{\mathrm{enc}}(\mathcal{M}_{\mathrm{mf}}),$$

thus preserving the interpretable representations of tissue composition and spatial organization learned at the WSI level.

Micro-scale branch

Micro-scale feature aggregation focuses on the top-K tiles $\{T^{(k)}\}_{k=1}^{K}$ selected according to attribution magnitude. For each tile $T^{(k)}$, we reuse the hand-crafted feature extractor to obtain a morphological feature vector

$$f\left(T^{(k)}\right) \in \mathbb{R}^{D}.$$

These tile-level feature vectors are averaged to produce a micro-scale feature vector

$$z_T = \frac{1}{K} \sum_{k=1}^{K} f\left(T^{(k)}\right),$$

where z_T summarizes the dominant micro-scale morphological patterns in the most prognostically informative regions.

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Feature fusion and prediction

The macro-scale (z_M) and micro-scale (z_T) vectors are concatenated into a unified multi-scale representation

$$h = [z_M, z_T],$$

which encodes both global tissue context and localized diagnostic patterns, enabling the model to correlate histological heterogeneity with clinical outcomes.

The MSFM prediction module extends the MFM design to process the multi-scale features h through an initial BN layer followed by three fully connected layers. The last layer outputs a scalar logit

$$\tilde{r} = g_{\text{pred}}(h),$$

which can be passed through a sigmoid to obtain a bounded network output

$$r = \sigma(\tilde{r}) \in (0, 1).$$

Here $g_{\text{pred}}(\cdot)$ denotes the MSFM prediction head. MSFM is trained with the same Cox partial likelihood loss $\mathcal{L}_{\text{cox}}$ as MFM.

The MSFM survival head is trained with the Cox partial-likelihood loss and includes an optional sigmoid mapping $r = \sigma(\tilde{r})$. For the reported patient-level survival analyses, the MorphX-derived risk score is the pre-sigmoid scalar $\tilde{r}$, interpreted as an unbounded, hazard-aligned prognostic index rather than a probability; higher values indicate higher modeled hazard.

5.6

Training protocol and hyperparameters

MSFM retains the MFM feature-extraction backbone and uses two hidden layers (64 and 32 nodes) for multi-scale feature integration. Sensitivity analyses evaluate top-K = 4,8,16,32,64 selected tiles. K = 16 is used for comparative analyses based on discovery-fold performance; the same K = 16 is used for all subsequent comparisons. External cohorts are not used to select K.

Training employs the Adam optimizer ($\beta = 0.9$) with malignancy-specific hyperparameters accounting for HCC and LUAD's differential tissue morphology. For HCC, MFM completes 15 training epochs (lr = 6.6×10^{-4}) before MSFM refinement through 10 additional epochs at a reduced learning rate (lr = 6.6×10^{-5}). For LUAD, MFM completes 40 training epochs (lr = 6.6×10^{-4}) followed by 20 MSFM fine-tuning epochs (lr = 9.9×10^{-5}). The MSFM architecture leverages transfer learning by initializing its macro-scale feature extraction backbone with pretrained weights from the optimized MFM. All preprocessing, feature extraction, and model training/inference are implemented in Python (RRID:SCR_008394). Deep learning models are implemented in PyTorch (RRID:SCR_018536).

Discovery cohorts use the one prespecified 10-fold partition described under Randomization and blinding. Each of the ten iterations uses 9-folds for training and the remaining TCGA fold for checkpoint selection and the reported fold-level discovery C-index; this quantity is therefore an internal selection metric and not a nested, optimism-corrected estimate of development performance. This is one reused 10-fold partition, not ten repeated random cross-validation runs. Each selected fold-specific model is then evaluated in KMMUFH or GDPH, which is excluded from adaptation, checkpoint selection, and hyperparameter selection. The same patient-to-fold assignments, endpoint, and selection protocol are used for eligible baseline comparisons.

For each cancer, the final MSFM is trained on the full discovery cohort and produces one continuous MorphX-derived prognostic risk score per patient. Higher scores indicate higher modeled hazard. The scores are used unchanged for the reported survival, clinical-integration, and feature-correlation analyses.

5.7

Baseline model comparisons

Baseline comparisons reuse the same prespecified TCGA patient-to-fold assignments and OS endpoint. For pretrained foundation models, including GigaPath, the pretrained encoder is kept fixed to extract tile-level representations rather than fine-tuned end-to-end for survival; a task-specific survival aggregation and prediction head is trained within each TCGA discovery fold using the Cox loss. MIL baselines are trained within the same TCGA folds using their specified architectures and input representations. The held-out TCGA fold is used for checkpoint selection, so the reported TCGA fold C-index is an internal selection metric. KMMUFH and GDPH are evaluated only after TCGA-only model selection and do not contribute to adaptation, checkpoint selection, or hyperparameter selection. Each comparator produces its own continuous prognostic score.

BEPH, pretrained on approximately 11 million unlabeled pathology patches, is included as a prognosis-oriented foundation-model comparator.[25] The downstream survival comparison reuses the same prespecified TCGA patient-to-fold assignments and OS endpoint, confines model selection to TCGA, evaluates each method using its own continuous risk score, and then applies the selected fold-specific models to KMMUFH or GDPH (Table S3; Data S2).

The MFM feature-composition ablation compares tissue type maps alone with complete morphological feature maps containing tissue type, tissue texture, and nuclear/cell descriptors (Table S2). Feature-family sensitivity models reuse the same prespecified TCGA patient-to-fold assignments, with checkpoint selection based on held-out TCGA internal-test C-index before evaluation in an external cohort that did not contribute to model selection or tuning. Evaluated inputs comprise tissue type maps, complete morphological feature maps, the indicated feature-family subsets, and the cohort-specific region-masking operations (Figures S2E-H, S5E-H; Data S2). Sensitivity to the number of selected tiles is shown separately in Figures S2C and S2D and Table S2. The complete-map values in Figures S2E–S2H provide descriptive references within the sensitivity-analysis workflow; primary MFM values are reported in Table S2.

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Morphology-based candidate screening and exploratory score construction

For each patient, we start from the integrated gradients attribution tensor $A \in \mathbb{R}^{w \times h \times (C+D)}$ computed for the input morphological feature map. Let $k \in \{1, \ldots, D\}$ index hand-crafted morphological features (e.g., GLCM-based texture or nuclear-shape features). We obtain an all-tissue feature-level attribution score for feature k in patient p by summing absolute attributions over all spatial locations (across all tissue types):

$$\alpha_k^{(p)} = \sum_{i=1}^{w} \sum_{j=1}^{h} |A_{i,j,k}|,$$

where $A_{i,j,k}$ denotes the attribution of feature k at position (i,j). For each patient in the discovery cohort, features are ranked by $\alpha_k^{(p)}$ and the top-10 features are recorded. Cohort-level all-tissue top features are then obtained by counting selection frequencies across patients, yielding a global top-10 list for HCC and LUAD separately.

For tissue type c, we define a binary mask $\mathbb{I}_{i,j,c}^{(p)}$ indicating whether tile (i,j) belongs to c. The tissue-type-specific attribution score of feature k in patient p is

$$\alpha_{k,c}^{(p)} = \sum_{i=1}^{w} \sum_{j=1}^{h} \mathbb{I}_{i,j,c}^{(p)} |A_{i,j,k}|.$$

For any feature k, we compute WSI-level mean values in two ways. (i) All-tissue (global) values average over all tissue tiles:

$$b_{k,\text{all}}^{(p)} = \frac{1}{N_{\text{all}}^{(p)}} \sum_{(i,j) \in \mathcal{T}_{\text{all}}^{(p)}} f_{i,j}^{(k)}.$$

(ii) Tissue-type-specific values average only tiles assigned to tissue c:

$$b_{k,c}^{(p)} = \frac{1}{N_c^{(p)}} \sum_{(i,j) \in \mathcal{T}_c^{(p)}} f_{i,j}^{(k)},$$

where the tile set for tissue type is used to calculate the corresponding WSI-level summary. For general tissue-specific summaries, an absent compartment is treated as missing. For the four LUAD candidates and their two-feature score, absent tissue-specific regions are assigned zero before discovery-cohort standardization. This convention preserves all patients but makes NEC DA, TUM DA, and STR DA partly reflect compartment presence as well as within-compartment texture; this convention is applied consistently in discovery and external score calculations.

For each tissue type, we scan its discovery-cohort tissue-specific top-10 feature list in rank order. For each feature, we compute the tissue-specific WSI-level value and determine a discovery cutoff by max-logrank, restricting candidate cut points to the 10th–90th percentiles to avoid extreme splits. The cutoff is transferred unchanged to the external cohort. We retain the first feature per tissue type that has nominal log rank $p < 0.05$ in discovery, a direction-consistent hazard ratio, and nominal log rank $p < 0.05$ in the external cohort. External outcomes therefore contribute to candidate retention. This procedure is an exploratory cross-cohort screen, not an untouched independent validation, and no multiplicity-adjusted confirmatory inference is claimed for the retained single candidates.

The combination analysis is restricted to four reported candidates per cancer: TUM MCC, NEC MCC, FIB LSR-SD, and STE Imc2 in HCC, and NEC DA, TUM DA, STR DA, and LYM DEN in LUAD. It is not an exhaustive search over the 160 morphology features. Candidate values are z-scaled using discovery-cohort parameters. A four-candidate discovery Cox model is first fitted as a conditional-association diagnostic rather than a significance-only prefilter, after which all 15 non-empty subsets are evaluated because correlated-feature coefficients can change across subsets. To obtain a multifeature score rather than another single-candidate model, eligible models are required to contain at least two features and to have nominal $p < 0.05$ for every included coefficient; eligible models are ranked by discovery partial AIC. No bootstrap or resampling stability criterion is applied. Discovery scaling and coefficients are applied externally without feature reselection or coefficient refitting. Exploratory complete-case multivariable Cox analyses of these scores use age, gender, source-recorded stage entered as categorical indicators, and the additional cohort-recorded fields listed in Table S5; these are sensitivity analyses rather than the grouped-stage primary Cox route (Figures S5I–S5L; Data S3).

LYM DEN is a WSI-level lymphocyte-density measure calculated from the cell_num channel in lymphocyte regions and normalized by total tissue area. An absent lymphocyte region is assigned zero before discovery-cohort standardization for score construction.

QUANTIFICATION AND STATISTICAL ANALYSIS

Survival performance

Model discrimination is evaluated using Harrell-style C-index, where higher predicted risk indicates shorter survival. Kaplan–Meier curves and log rank tests are used for visual risk stratification. For the primary MorphX risk-score analyses, discovery-derived cutoffs are transferred unchanged to external cohorts; the resulting group assignments are reported in Data S1. The separate tissue-specific candidate screen uses the cross-cohort retention rule described above.

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Cox regression and clinical integration

Cox proportional hazards models are used to estimate hazard ratios (HRs), 95% confidence intervals (CIs), and p values. Clinical variables are first evaluated in univariable models. Variables with univariable $p < 0.05$ enter the cohort-specific multivariable model together with the MorphX high-versus-low risk term (Figures 2C and 2D; Table S4). The primary models include grouped TNM stage in TCGA-LIHC; grouped BCLC stage and grouped differentiation in KMMUFH; grouped TNM stage in TCGA-LUAD; and grouped TNM stage, vascular invasion, differentiation, and smoking in GDPH. For the continuous-score analyses in Figures S2J and S2K, the within-cohort z-scaled final score enters without dichotomization. In both the primary high-versus-low and supplementary continuous-score clinical models, all missing included clinical covariates are handled by multiple imputation by chained equations (MICE; m = 20). For the supplementary continuous-score models, the imputation models include the analysis variables, the z-standardized MorphX score, OS event status, and the Nelson–Aalen cumulative-hazard estimate, with additional complete cohort-recorded predictors where available. Across imputed analyses, model coefficients and variances are combined using Rubin's rules. For the supplementary continuous-score models, clinical-only and combined C-indices are averaged across completed datasets. Ordinary Cox regression is used, except that the KMMUFH continuous-score model uses Firth bias-reduced Cox regression because of complete separation. Clinical-only and combined C-indices are fitted and evaluated within the same cohort and analysis population and are reported as apparent estimates without optimism correction. High-versus-low Cox results are reported in Figures 2C and 2D, Figure S2I, and Table S4; continuous-score results are reported in Figures S2J and S2K. Grouped TNM is coded as I–II versus III–IV in TCGA-LIHC, TCGA-LUAD, and GDPH, whereas KMMUFH uses grouped BCLC 0–1 versus 2–4.

For TCGA PFI, the endpoint source is the TCGA Clinical Data Resource (TCGA-CDR). PFI = 1 denotes a PFI event, PFI = 0 denotes censoring, and PFI.time is measured in days. The fixed OS-trained MorphX score is applied directly to PFI as a secondary endpoint in the same discovery patients; no PFI-specific retraining or tuning is performed, and the analysis is not an independent validation.

For the descriptive fixed-score 3-year OS comparison, death within 36 months defines the early-death group and documented survival beyond 36 months defines the longer-survival group; patients censored by 36 months are excluded. The prespecified OS-trained score is compared between these groups in the same discovery patients. Separately, the exploratory descriptor screen applies the same 36-month rule to OS and defines the PFI groups as progression by 36 months versus documented progression-free follow-up beyond 36 months, again excluding patients censored by 36 months. Within each endpoint, cancer, and cohort, the 160 prespecified slide-level descriptors are checked for variation, and non-invariant descriptors are compared using two-sided Mann–Whitney U tests. Rank-biserial effects are oriented so that positive values indicate higher descriptor values in the event-by-36-month group, and p values are adjusted by Benjamini–Hochberg within each screen. Discovery candidates are defined in TCGA at $q < 0.05$ and evaluated without reselection in the corresponding external cohort when matched endpoint data are available, with a second Benjamini–Hochberg correction across the transferred candidate tests. MorphX is not retrained and no 3-year classifier or subgroup-specific score is fitted.

Cohort characteristics and missing data

Cohort characteristics and non-missing denominators are summarized in Table S1. Data S1 provides the corresponding cohort-characteristics, model-variables, treatment-availability, and field-availability worksheets. KMMUFH contains three duplicate clinical keys in the source clinical table; one clinical row per modeled risk row is retained using non-negative survival time, completeness, and source order as tie-breakers.

Multiple testing and feature analyses

The tissue-specific candidate screen uses nominal log rank p values and is explicitly exploratory; Benjamini–Hochberg correction is not claimed for candidate retention. For risk–feature correlation, the MorphX-derived risk score is paired with each patient's WSI-level tissue-specific feature summary, Spearman correlations are computed separately within cohort, and Benjamini–Hochberg correction is applied within cohort. Because the score is constructed from the same morphology maps, these correlations are interpretability-consistency analyses rather than independent validation. Texture–nuclear-morphometry correlations and region-level nuclear-morphometry correlations are descriptive. To assess dataset-specific and overlapping features, tissue–feature entries are ranked independently in the discovery and external datasets within each cancer and compared at the exact tissue–feature level (Figure S5G; Data S3). For the cross-cancer comparison, each feature is represented by its maximum tissue-specific discovery selection frequency; features present in both cancers with an absolute frequency difference ≤ 0.20 define the four-feature shared group, and the remaining features define the 14-feature HCC-specific and 15-feature LUAD-specific groups. These discovery-defined groups are used in the feature-group sensitivity models (Figures S5F and S5H; Data S2 and S3). Because the groups are not size-matched and no threshold-sensitivity analysis is available, their performance comparison cannot isolate cancer specificity from feature-group size or threshold choice.

Matched TCGA bulk RNA-seq analysis

TCGA-LIHC and TCGA-LUAD discovery cases are matched to GDC Gene Expression Quantification files annotated as STAR - Counts and Primary Tumor. When multiple candidates exist, a deterministic selection rule prioritizes a Primary Tumor sample with suffix 01A and then applies deterministic sample/date/file identifiers. Protein-coding unstranded counts are assembled for 318 HCC and 282 LUAD cases with available files. Six targeted post hoc contrasts are analyzed using the median within the

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RNA-matched cases: overall MorphX risk in each cancer, HCC TUM MCC and FIB LSR-SD, and LUAD NEC DA and LYM DEN. These features are analyzed because they are the components retained by the stated two-feature score rule, not because other candidates were shown to lack molecular relevance. NEC MCC, STE Imc2, TUM DA, STR DA, and the remaining morphology features are not tested. Differential-expression *p* values are obtained from two-sided Welch t-tests on log2(CPM+1) values (unequal variances) and adjusted using the Benjamini-Hochberg procedure. Genes are ranked for 1,000-permutation Hallmark preranked GSEA (seed 20260701), and significant genes are used for GO Biological Process and KEGG legacy over-representation analysis using one-sided Fisher exact tests (greater alternative); the resulting over-representation *p* values are adjusted by Benjamini-Hochberg FDR within each analysis; minimum group size is 20. Figure S5N is a descriptive, non-exhaustive display of 10 representative pathways from the 50 Hallmark gene sets evaluated for each contrast. Cells labeled "ns" have FDR$\geq$0.05; cells labeled "NA" are FDR-significant source rows for which signed NES is unavailable and is not imputed. All 50 results per contrast are provided in Data S3 together with differential-expression and enrichment summaries. The number of FDR-significant genes is descriptive and is not used to rank features or infer a stronger biological effect across contrasts. These analyses provide targeted cohort-scale molecular context rather than spatial, causal, or independent biomarker validation.

Software

Statistical analyses are performed in Python 3.10.9 using numpy 1.23.5, pandas 1.5.3, statsmodels 0.13.5, openpyxl 3.0.10, matplotlib 3.7.0, scipy 1.10.0, scikit-learn 1.2.1, PyTorch 1.12.1, and Pillow 9.4.0.